Institutions → Outcomes

Which institutional bundles explain fiscal, health, education, and complexity outcomes?

Institutional Complexity Project

2026-04-24

Contents

Motivation

The institutional-complexity analysis describes *how* institutional features cluster. This companion document asks the next question: **which institutional bundles predict real-world outcomes?** For each of ten cross-country outcomes (fiscal, health, education, economic complexity) we fit a flexible gradient-boosted model on the full V-Dem feature set and use SHAP values to surface (1) which individual institutional features matter most and (2) which *pairs* of features interact most strongly. Shallow trees (`max_depth = 3`) restrict the model to at-most-three-way interactions — the "bundle" hypothesis says pairwise and low-order interactions should dominate over single-feature main effects.

This is a descriptive exercise, not a causal one. V-Dem scores and fiscal/health/ education outcomes are co-determined over long horizons; no design here identifies a causal effect of institutions on outcomes.

Data

Rebuild the institutional Mcp matrix

We reproduce the V-Dem preprocessing from the main analysis: the most recent year with at least 150-country coverage, all C-type components with good coverage, min-max normalised to [0, 1]. See `institutional_complexity.Rmd` for detail.

```
## V-Dem year: 2025
```

```
## Mcp: 179 countries × 185 features
```

Load outcome variables from the macro parquet

Ten outcomes across four buckets. V-Dem's latest-coverage year runs ahead of several macro sources (HCI in particular is released only every 2–3 years), so for each country × outcome we take the **most recent non-NA value within the six years ending at `latest_year`**. This trades a little temporal precision for usable cross-country coverage. The "year used" is reported per outcome. `qog_bl_asymf` (Barro-Lee avg years of schooling) was dropped — its latest non-NA observation in this vintage is 2015, too stale to pair with a V-Dem 2025 snapshot.

Table 1: Outcome variables. `log` indicates a natural-log transform was applied before modeling. Debt uses log1p because small-debt observations are near zero.

column	bucket	label	transform	source
weo__ggr__ngdp	Fiscal	Revenue % GDP	none	macro
imf_tax_inc	Fiscal	Income tax % GDP	none	imf_world
wdi__sp__dyn__le00__in	Health	Life expectancy	none	macro
wdi__sp__dyn__imrt__in	Health	Infant mortality (log)	log	macro
wdi__sh__sta__mmrt	Health	Maternal mortality (log)	log	macro
wdi__hd__hci__ovrl	Education	Human Capital Index	none	macro
wdi__se__sec__enrr	Education	Secondary enrollment %	none	macro
atl__sitc__eci	Complexity	Economic Complexity (ECI)	none	macro
wdi__ne__exp__gnfs__zs	Trade	Exports % GDP	none	macro

Join to Mcp

```
## Rows in analysis frame: 172
```

```
## Countries in Mcp but missing macro match: 7
```

Table 2: Per-outcome country coverage and the range of years used in the joined analysis frame.

bucket	label	column	transform	n	year_range
Fiscal	Revenue % GDP	weo__ggr__ngdp	none	170	2021–2025 (median 2025)
Fiscal	Income tax % GDP	imf__tax__inc	none	166	2023–2024 (median 2024)
Health	Life expectancy	wdi__sp__dyn__le00__in	none	171	2023–2023 (median 2023)
Health	Infant mortality (log)	wdi__sp__dyn__imrt__in	log	170	2023–2023 (median 2023)
Health	Maternal mortality (log)	wdi__sh__sta__mmrt	log	170	2023–2023 (median 2023)
Education	Human Capital Index	wdi__hd__hci__ovrl	none	156	2020–2020 (median 2020)
Education	Secondary enrollment %	wdi__se__sec__enrr	none	144	2020–2023 (median 2022)
Complexity	Economic Complexity (ECI)	atl__sitc__eci	none	172	2023–2023 (median 2023)

bucket	label	column	transform	n	year_range
Trade	Exports % GDP	wdi_ne_exp_gnfs_zs	none	155	2022–2024 (median 2024)

Modeling

Pipeline

For each outcome we fit XGBoost at each of four tree depths, `max_depth in {3, 4, 5, 6}`. Shallower trees cap the interaction order the model can learn — depth 3 allows up to three-way interactions, depth 6 up to six-way. Out-of-sample R^2 is computed from 5-fold CV predictions at each depth; comparing the R^2 curve across depths per outcome is the "how much does bundle *size* matter" diagnostic.

Other hyperparameters are held fixed: `eta = 0.05`, `subsample = 0.8`, `colsample_bytree = 0.8`, `min_child_weight = 5` (the small N makes this last one important — it prevents leaves from specializing on one or two countries).

Depth sweep

```
## Fitting: weo_ggr_ngdp @ depth 3

## Fitting: weo_ggr_ngdp @ depth 4

## Fitting: weo_ggr_ngdp @ depth 5

## Fitting: weo_ggr_ngdp @ depth 6

## Fitting: imf_tax_inc @ depth 3

## Fitting: imf_tax_inc @ depth 4

## Fitting: imf_tax_inc @ depth 5

## Fitting: imf_tax_inc @ depth 6

## Fitting: wdi_sp_dyn_le00_in @ depth 3

## Fitting: wdi_sp_dyn_le00_in @ depth 4

## Fitting: wdi_sp_dyn_le00_in @ depth 5

## Fitting: wdi_sp_dyn_le00_in @ depth 6

## Fitting: wdi_sp_dyn_imrt_in @ depth 3

## Fitting: wdi_sp_dyn_imrt_in @ depth 4
```

```
## Fitting: wdi_sp_dyn_imrt_in @ depth 5

## Fitting: wdi_sp_dyn_imrt_in @ depth 6

## Fitting: wdi_sh_sta_mmrt @ depth 3

## Fitting: wdi_sh_sta_mmrt @ depth 4

## Fitting: wdi_sh_sta_mmrt @ depth 5

## Fitting: wdi_sh_sta_mmrt @ depth 6

## Fitting: wdi_hd_hci_ovrl @ depth 3

## Fitting: wdi_hd_hci_ovrl @ depth 4

## Fitting: wdi_hd_hci_ovrl @ depth 5

## Fitting: wdi_hd_hci_ovrl @ depth 6

## Fitting: wdi_se_sec_enrr @ depth 3

## Fitting: wdi_se_sec_enrr @ depth 4

## Fitting: wdi_se_sec_enrr @ depth 5

## Fitting: wdi_se_sec_enrr @ depth 6

## Fitting: atl_sitc_eci @ depth 3

## Fitting: atl_sitc_eci @ depth 4

## Fitting: atl_sitc_eci @ depth 5

## Fitting: atl_sitc_eci @ depth 6

## Fitting: wdi_ne_exp_gnfs_zs @ depth 3

## Fitting: wdi_ne_exp_gnfs_zs @ depth 4

## Fitting: wdi_ne_exp_gnfs_zs @ depth 5

## Fitting: wdi_ne_exp_gnfs_zs @ depth 6
```

Table 3: Out-of-sample R^2 by outcome and max_depth. Each column is a fit with that interaction cap.

bucket	label	d3	d4	d5	d6
Complexity	Economic Complexity (ECI)	0.450	0.447	0.439	0.443
Education	Human Capital Index	0.771	0.762	0.762	0.765
Education	Secondary enrollment %	0.513	0.517	0.499	0.510
Fiscal	Income tax % GDP	0.563	0.579	0.581	0.572
Fiscal	Revenue % GDP	0.352	0.362	0.371	0.355
Health	Infant mortality (log)	0.703	0.702	0.710	0.708
Health	Maternal mortality (log)	0.684	0.669	0.663	0.658
Health	Life expectancy	0.629	0.609	0.594	0.613
Trade	Exports % GDP	0.108	0.103	0.143	0.155

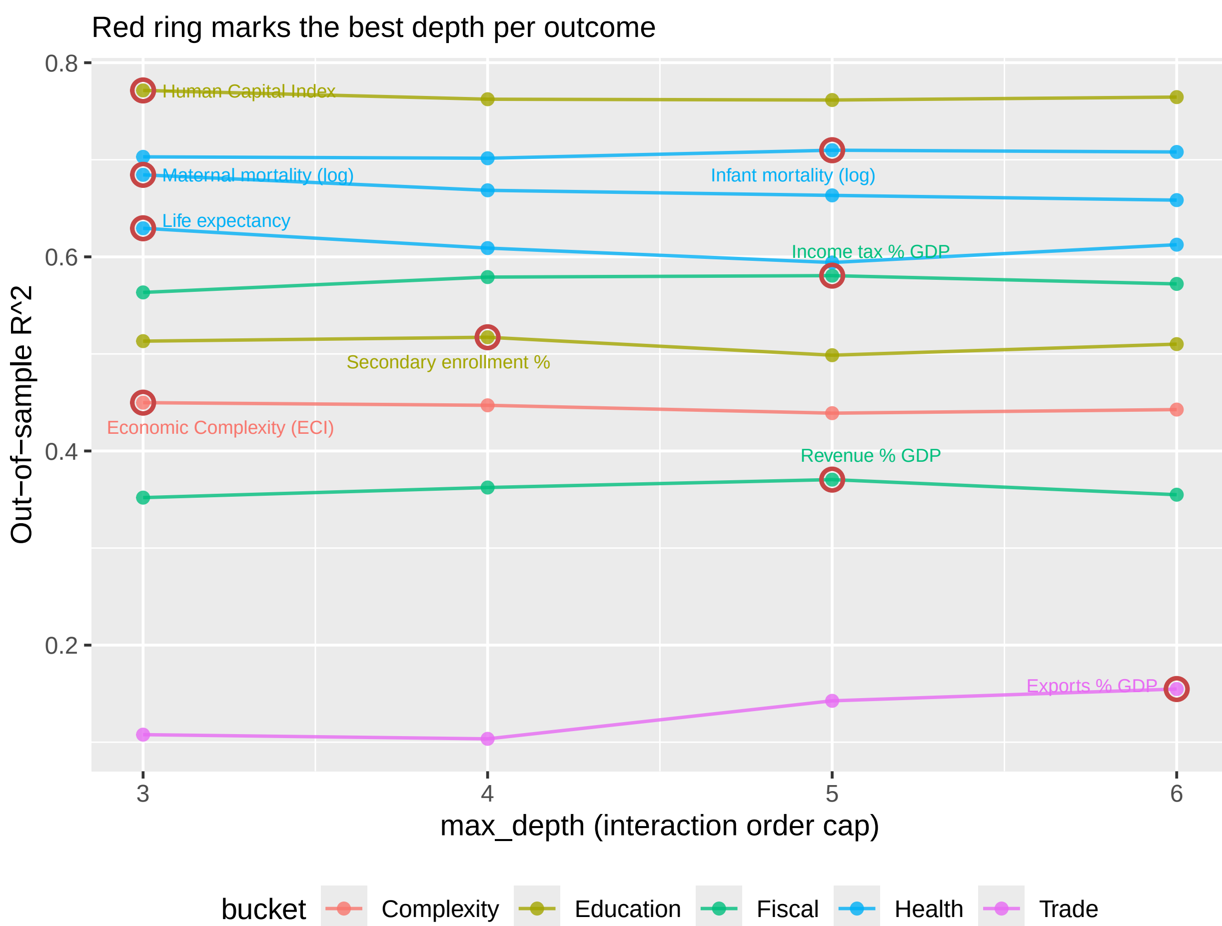


Table 4: Best depth per outcome. `Gain` = best R^2 minus R^2 at depth 3 — i.e. how much moving beyond three-way bundles actually helps.

Bucket	Outcome	Best depth	Best R^2	R^2 @ d3	Gain
Complexity	Economic Complexity (ECI)	3	0.450	0.450	0.000

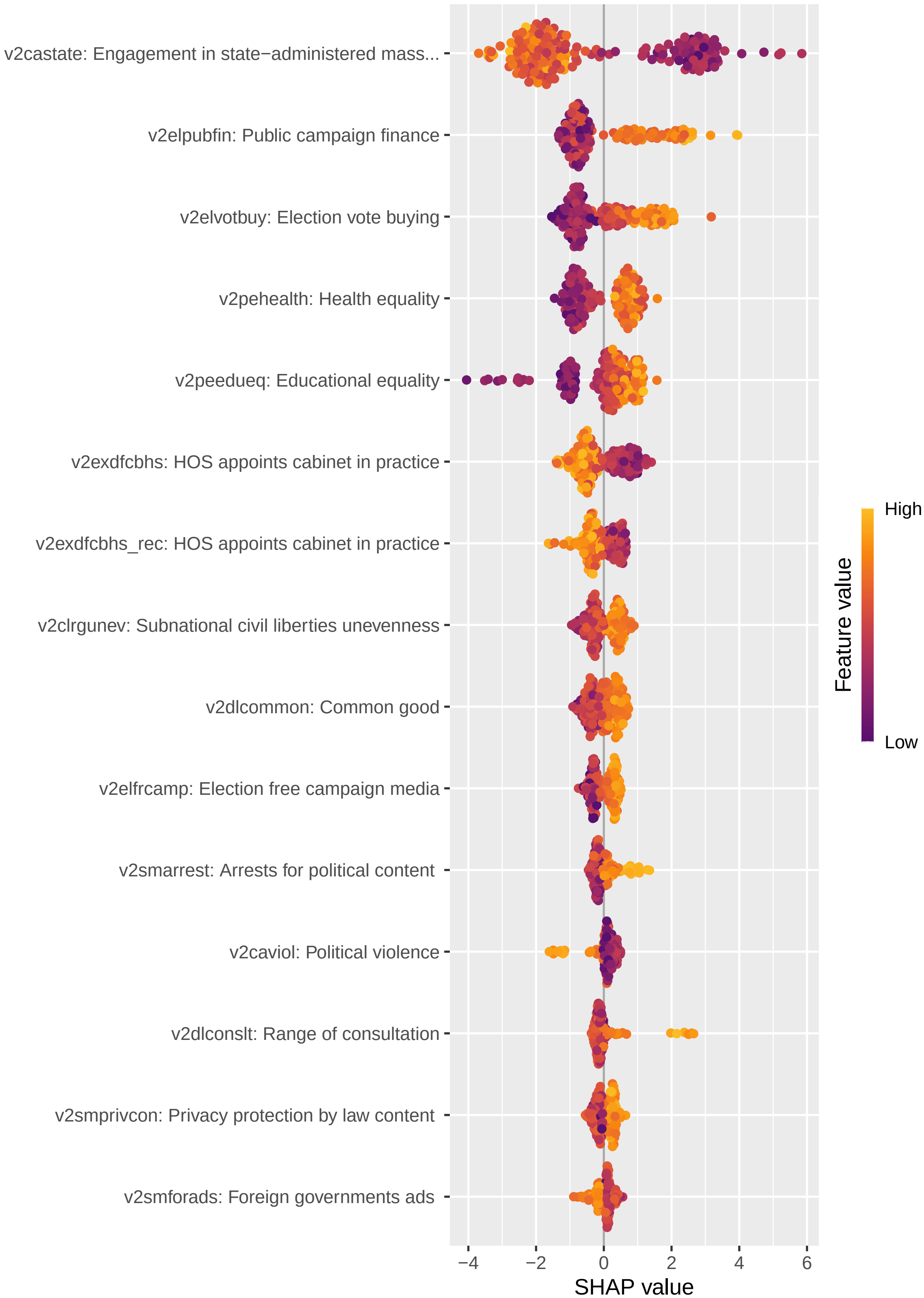
Bucket	Outcome	Best depth	Best R^2	R^2 @ d3	Gain
Education	Secondary enrollment %	4	0.517	0.513	0.004
Education	Human Capital Index	3	0.771	0.771	0.000
Fiscal	Revenue % GDP	5	0.371	0.352	0.019
Fiscal	Income tax % GDP	5	0.581	0.563	0.018
Health	Infant mortality (log)	5	0.710	0.703	0.007
Health	Life expectancy	3	0.629	0.629	0.000
Health	Maternal mortality (log)	3	0.684	0.684	0.000
Trade	Exports % GDP	6	0.155	0.108	0.047

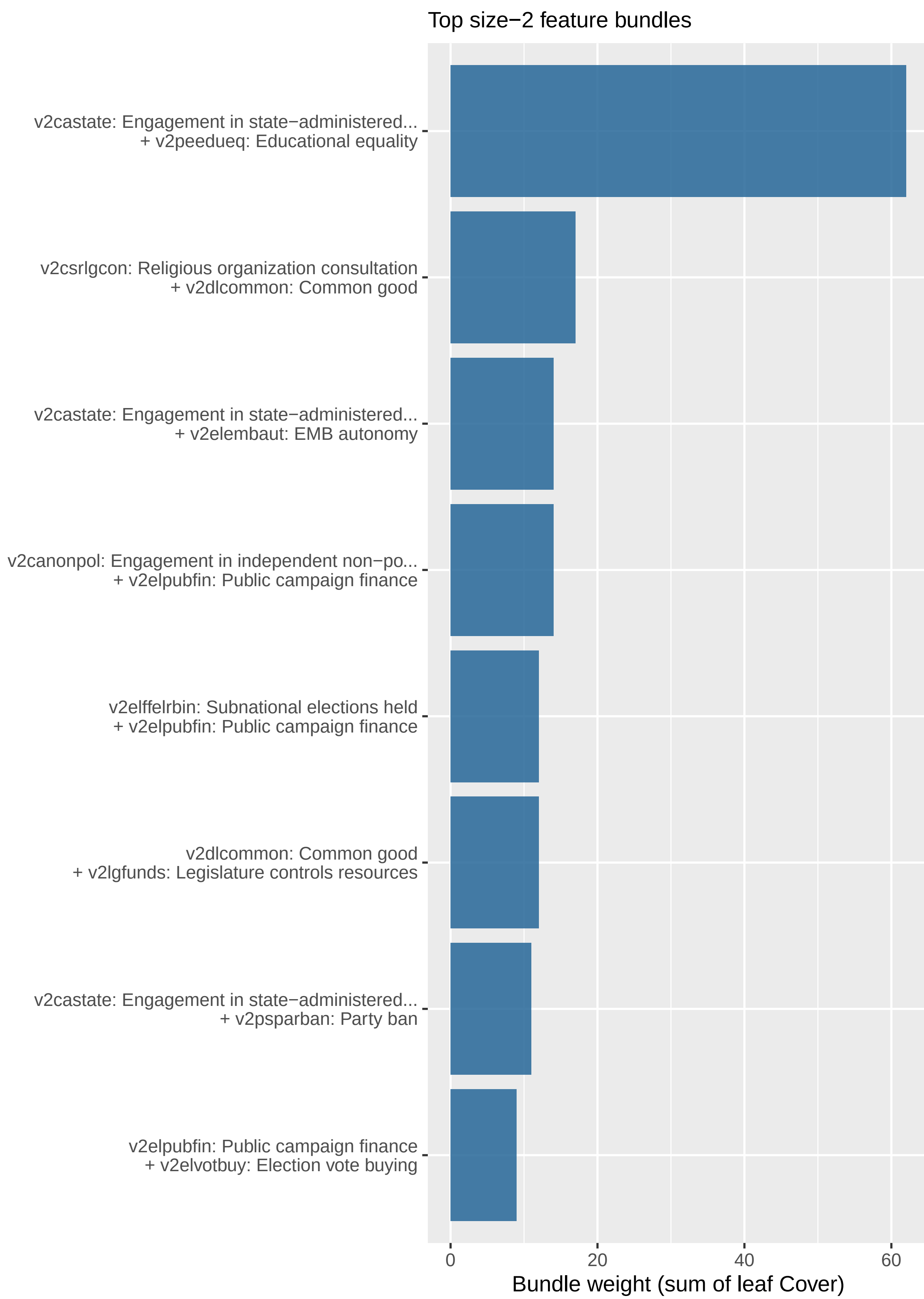
Per-outcome detail (at best depth)

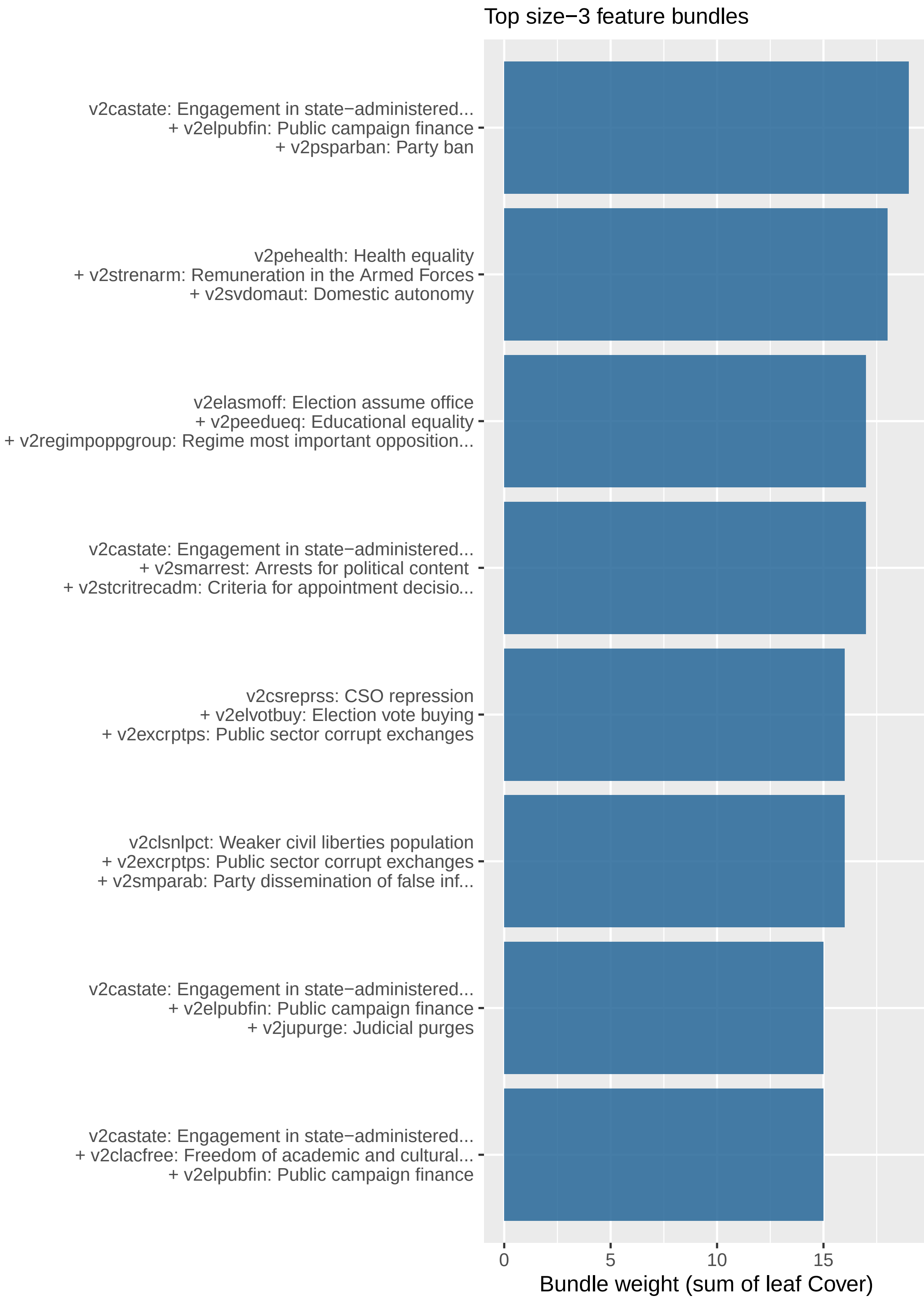
Each outcome's section below uses the model at its best-performing depth. We show (1) a SHAP beeswarm of the top individual features — i.e. where each feature falls across the country distribution and how it pushes predictions — and (2) the top **feature bundles** of each size, extracted by walking the tree leaves. Bundle weight = sum of leaf `Cover` for all leaves whose path uses exactly that set of features; high-weight bundles are conjunctions of features the model relies on repeatedly across many countries.

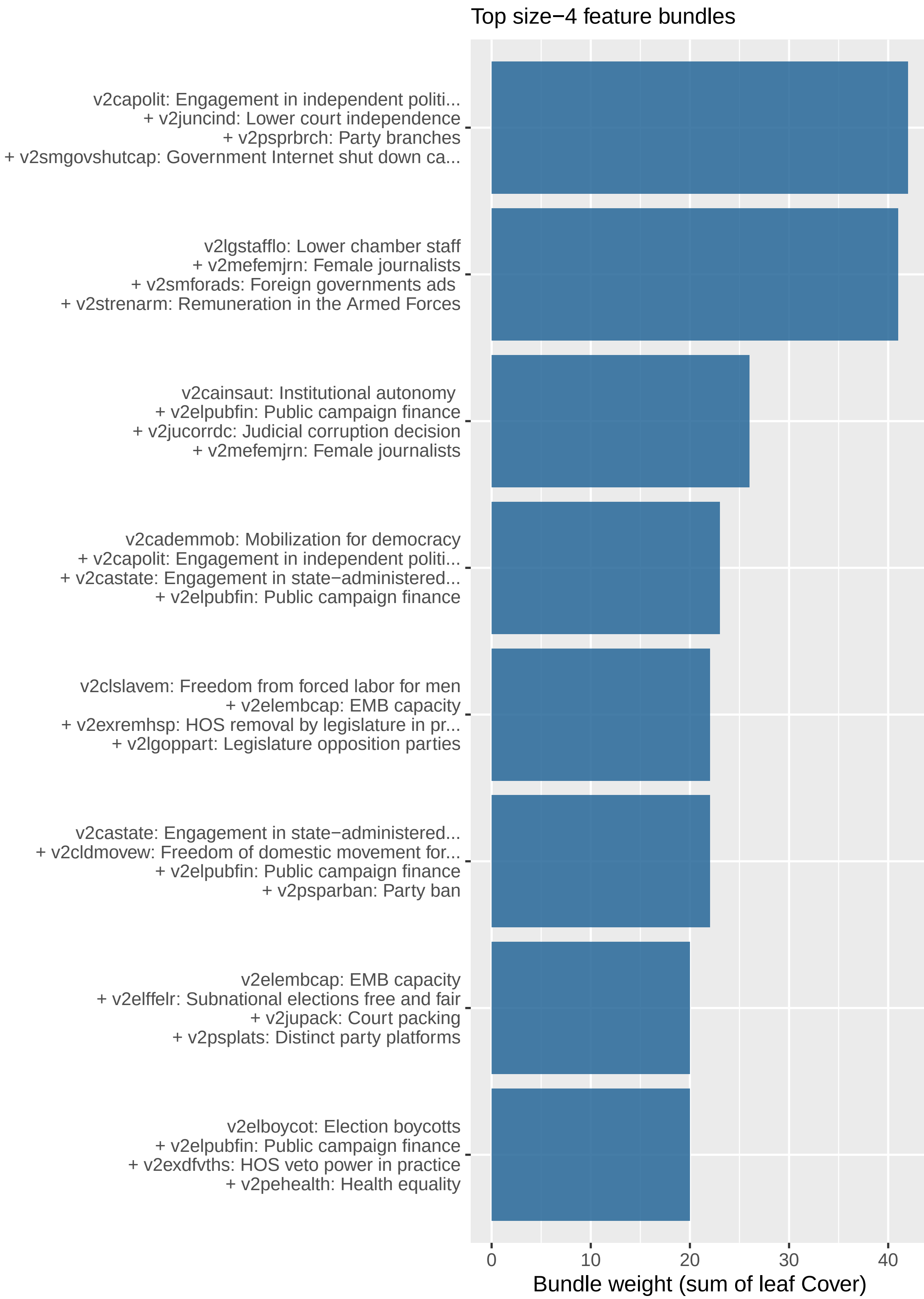
Revenue % GDP

N = 170 | best depth = 5 | OOS R^2 = 0.371 | nrounds = 64


v2castate: Engagement in state−administered mass...
v2elpubfin: Public campaign finance
v2elvotbuy: Election vote buying
v2pehealth: Health equality
v2peedueq: Educational equality
v2exdfcbhs: HOS appoints cabinet in practice
v2exdfcbhs_rec: HOS appoints cabinet in practice
v2clrgunev: Subnational civil liberties unevenness
v2dlcommon: Common good
v2elfrcamp: Election free campaign media
v2smarrest: Arrests for political content
v2caviol: Political violence
v2dlconslt: Range of consultation
v2smprivcon: Privacy protection by law content
v2smforads: Foreign governments ads
−4
−2
0
2
4
6
SHAP value
High
Feature value
Low


Top size−2 feature bundles
v2castate: Engagement in state−administered...
+ v2peedueq: Educational equality
v2csrlgcon: Religious organization consultation
+ v2dlcommon: Common good
v2castate: Engagement in state−administered...
+ v2elembaut: EMB autonomy
v2canonpol: Engagement in independent non−po...
+ v2elpubfin: Public campaign finance
v2elffelrbin: Subnational elections held
+ v2elpubfin: Public campaign finance
v2dlcommon: Common good
+ v2lgfunds: Legislature controls resources
v2castate: Engagement in state−administered...
+ v2psparban: Party ban
v2elpubfin: Public campaign finance
+ v2elvotbuy: Election vote buying
0
20
40
60
Bundle weight (sum of leaf Cover)

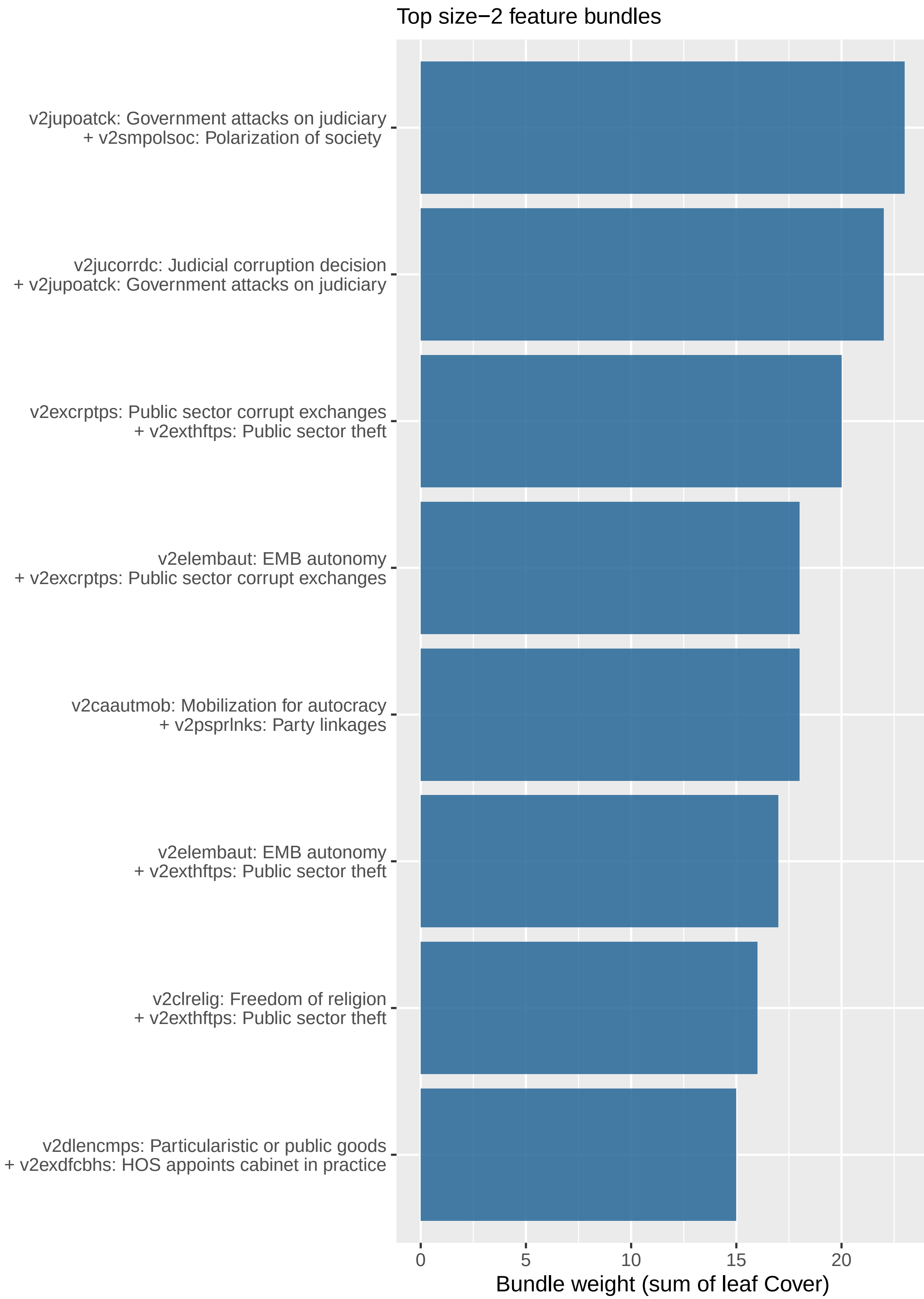

Top size−3 feature bundles
v2castate: Engagement in state−administered...
+ v2elpubfin: Public campaign finance
+ v2psparban: Party ban
v2pehealth: Health equality
+ v2strenarm: Remuneration in the Armed Forces
+ v2svdomaut: Domestic autonomy
v2elasmoff: Election assume office
+ v2peedueq: Educational equality
+ v2regimpoppgroup: Regime most important opposition...
v2castate: Engagement in state−administered...
+ v2smarrest: Arrests for political content
+ v2stcritrecadm: Criteria for appointment decisio...
v2csreprss: CSO repression
+ v2elvotbuy: Election vote buying
+ v2excrptps: Public sector corrupt exchanges
v2clsnlpct: Weaker civil liberties population
+ v2excrptps: Public sector corrupt exchanges
+ v2smparab: Party dissemination of false inf...
v2castate: Engagement in state−administered...
+ v2elpubfin: Public campaign finance
+ v2jupurge: Judicial purges
v2castate: Engagement in state−administered...
+ v2clacfree: Freedom of academic and cultural...
+ v2elpubfin: Public campaign finance
0
5
10
15
Bundle weight (sum of leaf Cover)


Top size−4 feature bundles
v2capolit: Engagement in independent politi...
+ v2juncind: Lower court independence
+ v2psprbrch: Party branches
+ v2smgovshutcap: Government Internet shut down ca...
v2lgstafflo: Lower chamber staff
+ v2mefemjrn: Female journalists
+ v2smforads: Foreign governments ads
+ v2strenarm: Remuneration in the Armed Forces
v2cainsaut: Institutional autonomy
+ v2elpubfin: Public campaign finance
+ v2jucorrdc: Judicial corruption decision
+ v2mefemjrn: Female journalists
v2cademmob: Mobilization for democracy
+ v2capolit: Engagement in independent politi...
+ v2castate: Engagement in state−administered...
+ v2elpubfin: Public campaign finance
v2clslavem: Freedom from forced labor for men
+ v2elembcap: EMB capacity
+ v2exremhsp: HOS removal by legislature in pr...
+ v2lgoppart: Legislature opposition parties
v2castate: Engagement in state−administered...
+ v2cldmovew: Freedom of domestic movement for...
+ v2elpubfin: Public campaign finance
+ v2psparban: Party ban
v2elembcap: EMB capacity
+ v2elffelr: Subnational elections free and fair
+ v2jupack: Court packing
+ v2psplats: Distinct party platforms
v2elboycot: Election boycotts
+ v2elpubfin: Public campaign finance
+ v2exdfvths: HOS veto power in practice
+ v2pehealth: Health equality
0
10
20
30
40
Bundle weight (sum of leaf Cover)

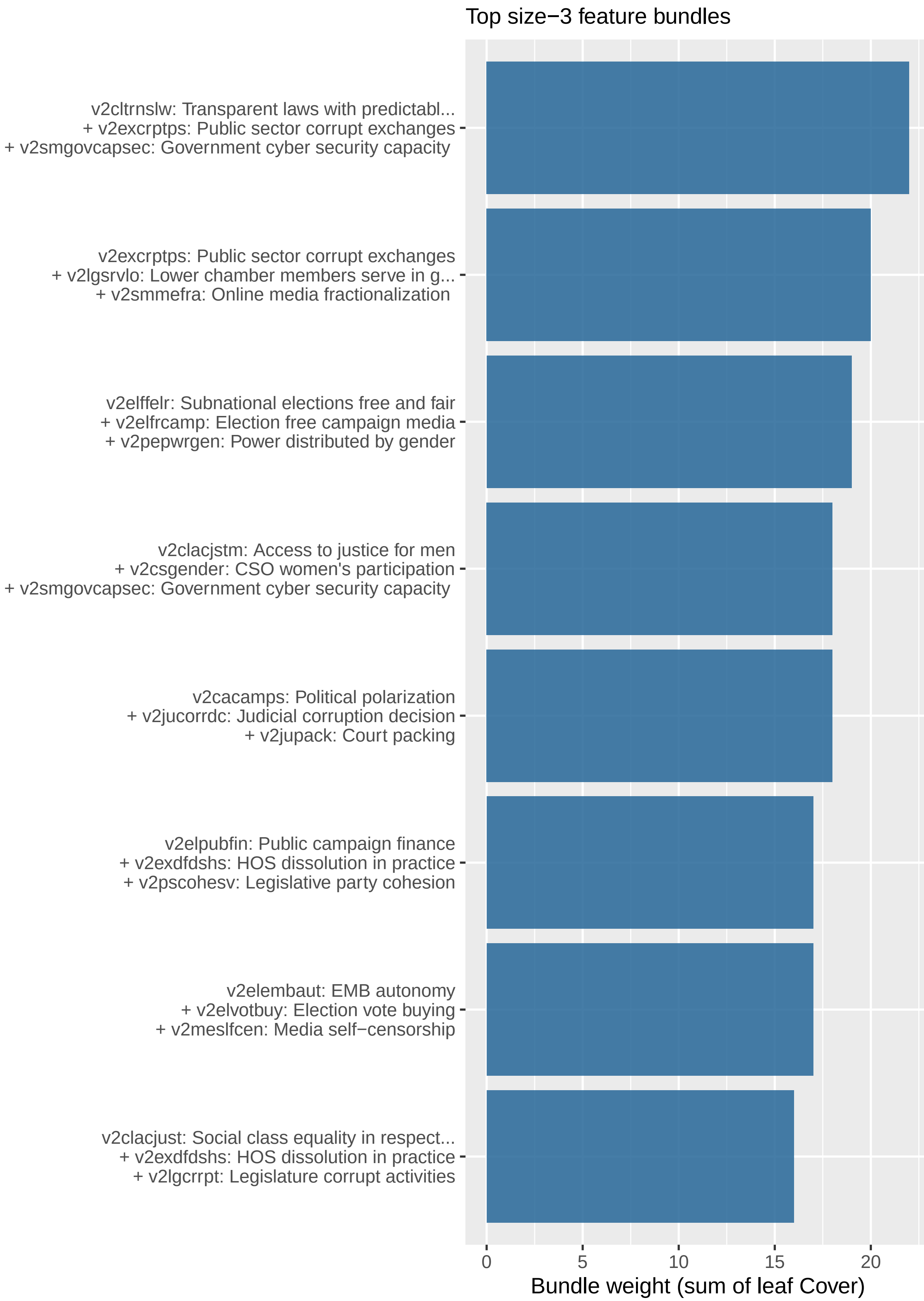
Income tax % GDP

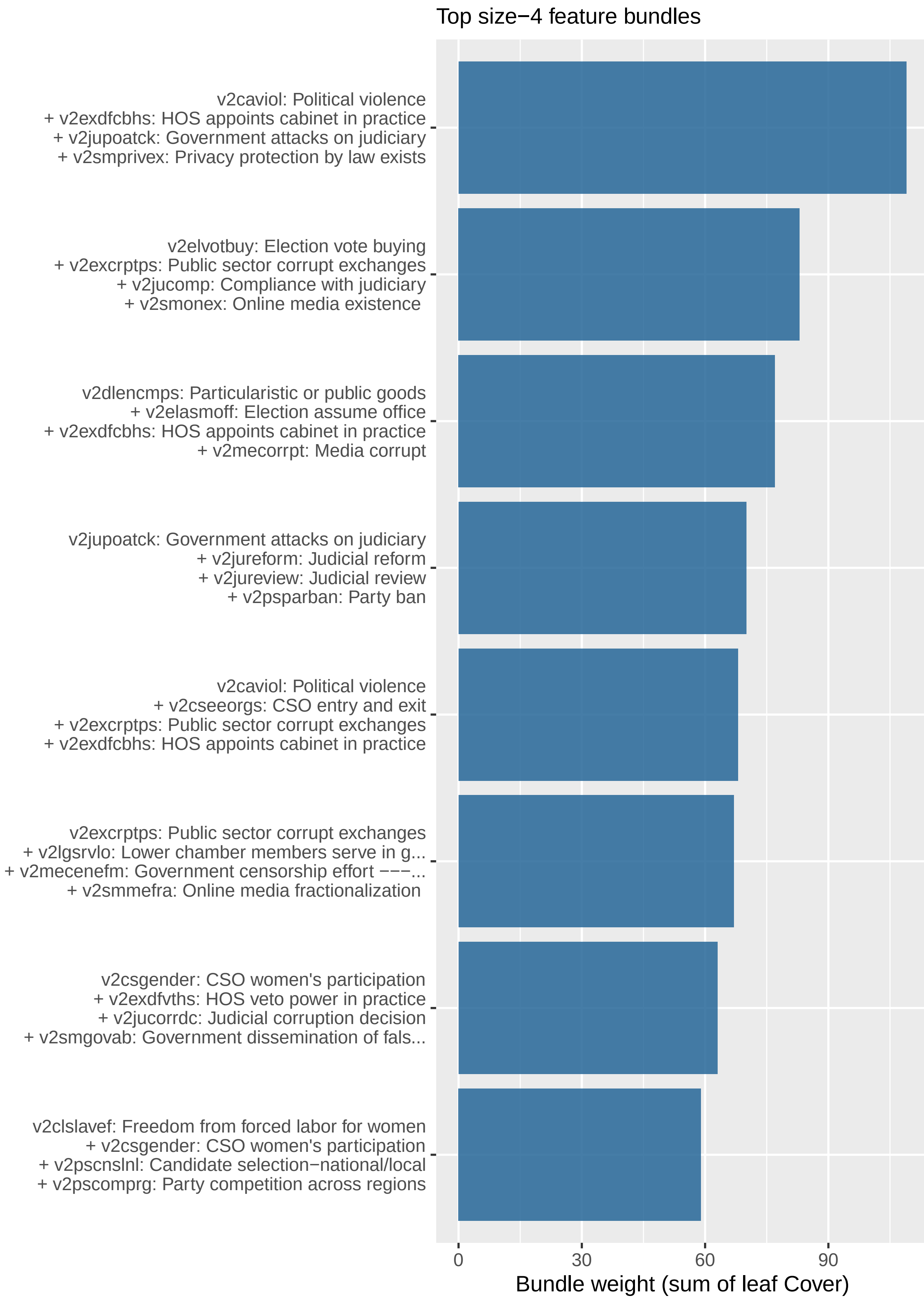
N = 166 / best depth = 5 / OOS R^2 = 0.581 / nrounds = 135


v2excrptps: Public sector corrupt exchanges
v2exthftps: Public sector theft
v2elffelr: Subnational elections free and fair
v2mefemjrn: Female journalists
v2psprlnks: Party linkages
v2dlcommon: Common good
v2jucorrdc: Judicial corruption decision
v2stfisccap: State fiscal source of revenue
v2clrelig: Freedom of religion
v2elembaut: EMB autonomy
v2elvotbuy: Election vote buying
v2lglegplo: Lower chamber legislates in practice
v2catrauni: Engagement in independent trade unions
v2elembcap: EMB capacity
v2lgcrrpt: Legislature corrupt activities
0
1
2
3
SHAP value
High
Feature value
Low



Top size−2 feature bundles
v2jupoatck: Government attacks on judiciary
+ v2smpolsoc: Polarization of society
v2jucorrdc: Judicial corruption decision
+ v2jupoatck: Government attacks on judiciary
v2excrptps: Public sector corrupt exchanges
+ v2exthftps: Public sector theft
v2elembaut: EMB autonomy
+ v2excrptps: Public sector corrupt exchanges
v2caautmob: Mobilization for autocracy
+ v2psprlnks: Party linkages
v2elembaut: EMB autonomy
+ v2exthftps: Public sector theft
v2clrelig: Freedom of religion
+ v2exthftps: Public sector theft
v2dlencmps: Particularistic or public goods
+ v2exdfcbhs: HOS appoints cabinet in practice
0
5
10
15
20
Bundle weight (sum of leaf Cover)


Top size−3 feature bundles
v2cltrnslw: Transparent laws with predictabl...
+ v2excrptps: Public sector corrupt exchanges
+ v2smgovcapsec: Government cyber security capacity
v2excrptps: Public sector corrupt exchanges
+ v2lgsrvlo: Lower chamber members serve in g...
+ v2smmefra: Online media fractionalization
v2elffelr: Subnational elections free and fair
+ v2elfrcamp: Election free campaign media
+ v2pepwrgen: Power distributed by gender
v2clacjstm: Access to justice for men
+ v2csgender: CSO women's participation
+ v2smgovcapsec: Government cyber security capacity
v2cacamps: Political polarization
+ v2jucorrdc: Judicial corruption decision
+ v2jupack: Court packing
v2elpubfin: Public campaign finance
+ v2exdfdshs: HOS dissolution in practice
+ v2pscohesv: Legislative party cohesion
v2elembaut: EMB autonomy
+ v2elvotbuy: Election vote buying
+ v2meslfcen: Media self−censorship
v2clacjust: Social class equality in respect...
+ v2exdfdshs: HOS dissolution in practice
+ v2lgcrrpt: Legislature corrupt activities
0
5
10
15
20
Bundle weight (sum of leaf Cover)

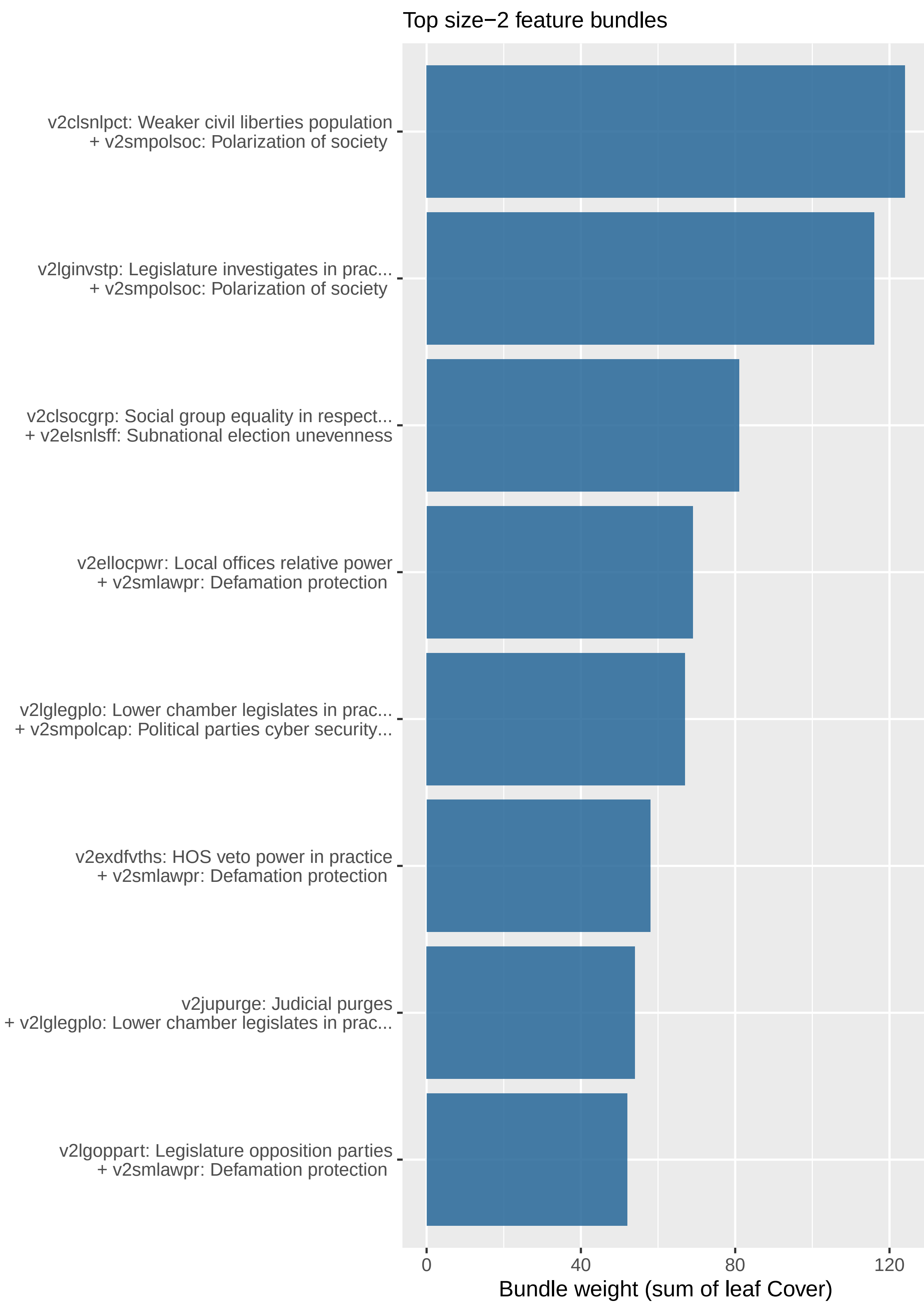


Top size−4 feature bundles
v2caviol: Political violence
+ v2exdfcbhs: HOS appoints cabinet in practice
+ v2jupoatck: Government attacks on judiciary
+ v2smprivex: Privacy protection by law exists
v2elvotbuy: Election vote buying
+ v2excrptps: Public sector corrupt exchanges
+ v2jucomp: Compliance with judiciary
+ v2smonex: Online media existence
v2dlencmps: Particularistic or public goods
+ v2elasmoff: Election assume office
+ v2exdfcbhs: HOS appoints cabinet in practice
+ v2mecorrpt: Media corrupt
v2jupoatck: Government attacks on judiciary
+ v2jureform: Judicial reform
+ v2jureview: Judicial review
+ v2psparban: Party ban
v2caviol: Political violence
+ v2cseeorgs: CSO entry and exit
+ v2excrptps: Public sector corrupt exchanges
+ v2exdfcbhs: HOS appoints cabinet in practice
v2excrptps: Public sector corrupt exchanges
+ v2lgsrvlo: Lower chamber members serve in g...
+ v2mecenefm: Government censorship effort −−−...
+ v2smmefra: Online media fractionalization
v2csgender: CSO women's participation
+ v2exdfvths: HOS veto power in practice
+ v2jucorrdc: Judicial corruption decision
+ v2smgovab: Government dissemination of fals...
v2clslavef: Freedom from forced labor for women
+ v2csgender: CSO women's participation
+ v2pscnslnl: Candidate selection−national/local
+ v2pscomprg: Party competition across regions
0
30
60
90
Bundle weight (sum of leaf Cover)

Life expectancy

N = 171 / best depth = 3 / OOS R^2 = 0.629 / nrounds = 206


v2pehealth: Health equality
v2smonex: Online media existence
v2exthftps: Public sector theft
v2clprptyw: Property rights for women
v2smpolcap: Political parties cyber security capa...
v2elembcap: EMB capacity
v2elvotbuy: Election vote buying
v2catrauni: Engagement in independent trade unions
v2castate: Engagement in state−administered mass...
v2mefemjrn: Female journalists
v2peedueq: Educational equality
v2exbribe: Executive bribery and corrupt exchanges
v2regoppgroupssize: Regime opposition groups size
v2excrptps: Public sector corrupt exchanges
v2elffelr: Subnational elections free and fair
−2
0
2
SHAP value
Feature value
High
Low

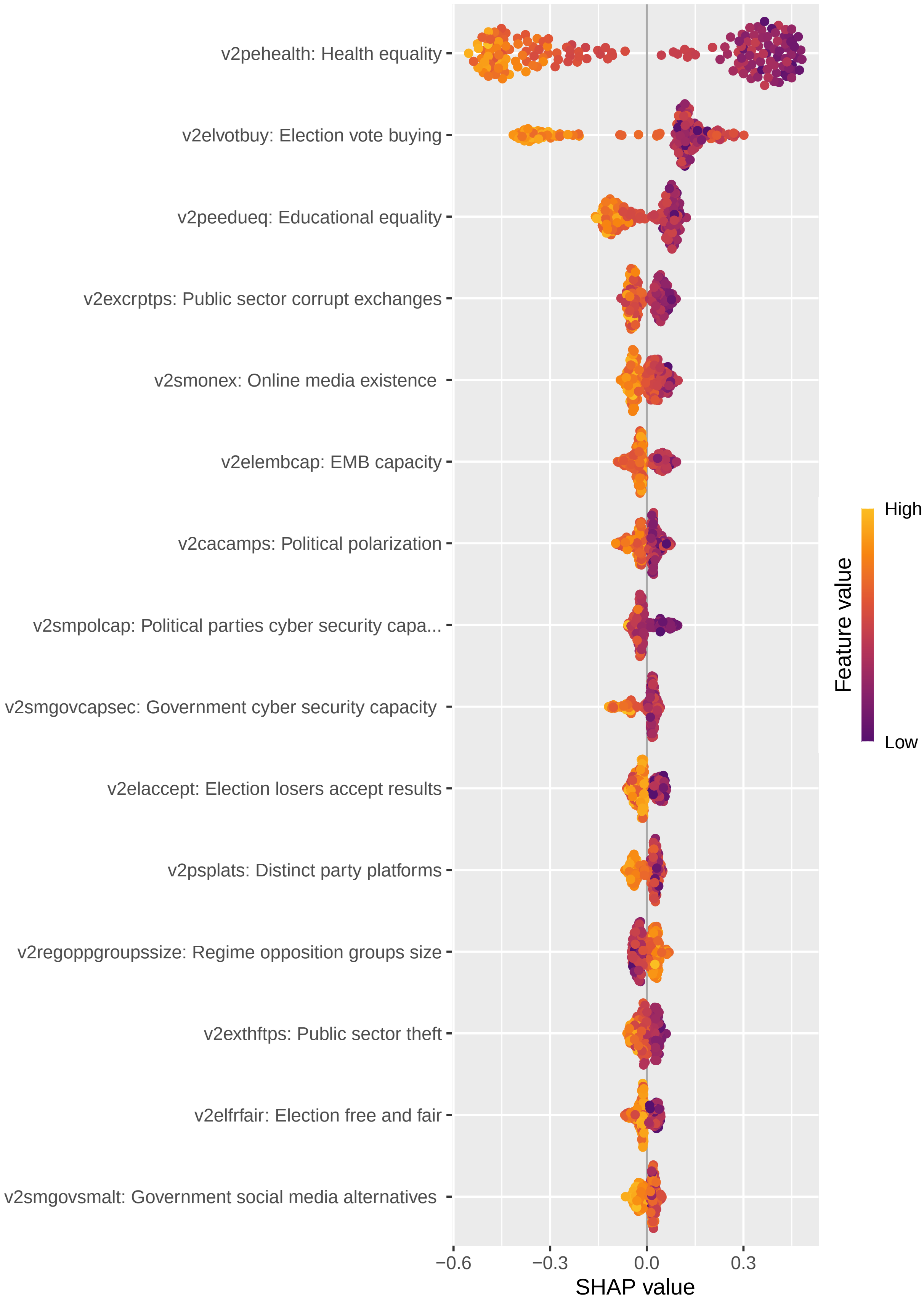


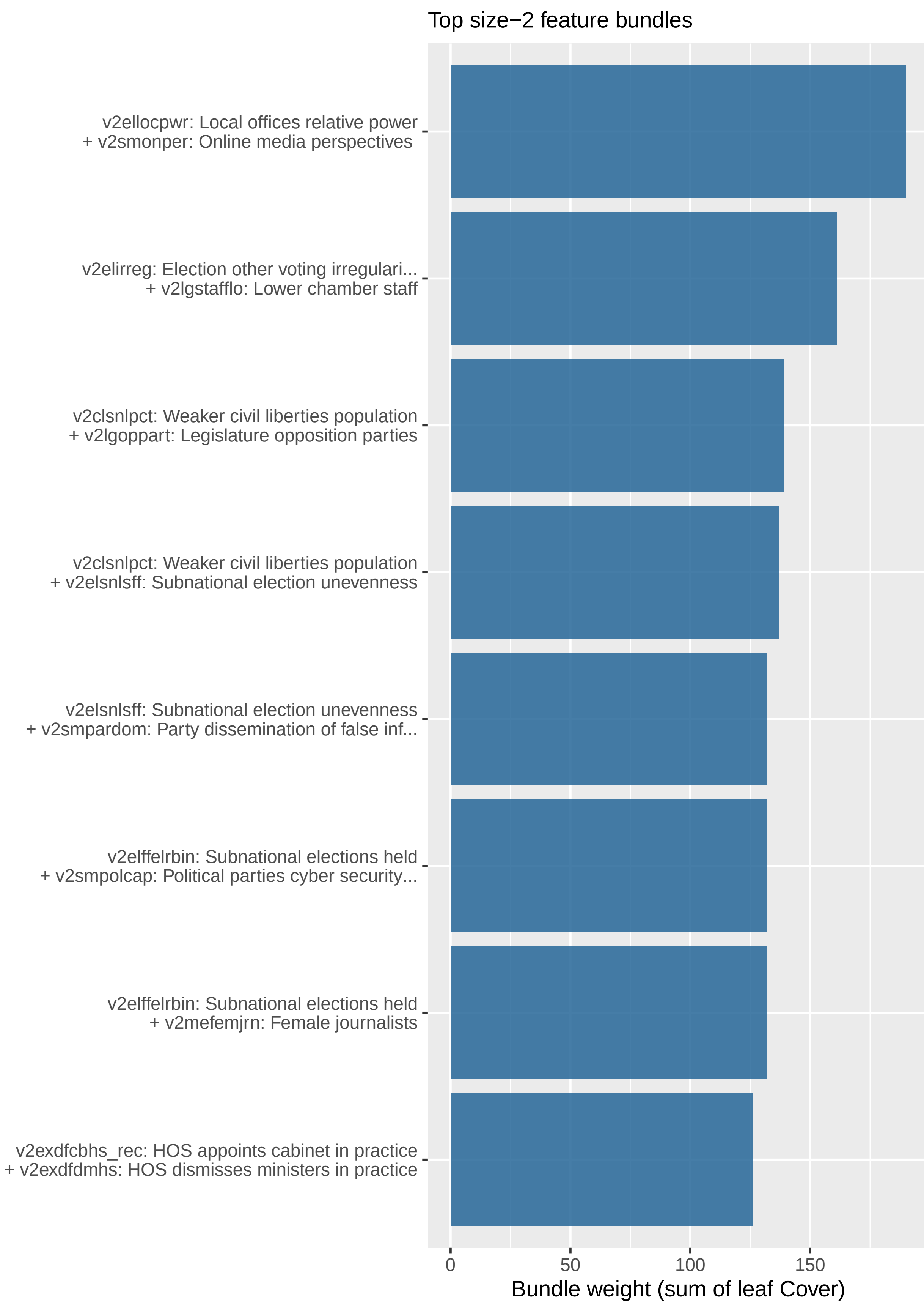
Top size−2 feature bundles
v2clsnlpct: Weaker civil liberties population
+ v2smpolsoc: Polarization of society
v2lginvstp: Legislature investigates in prac...
+ v2smpolsoc: Polarization of society
v2clsocgrp: Social group equality in respect...
+ v2elsnlsff: Subnational election unevenness
v2ellocpwr: Local offices relative power
+ v2smlawpr: Defamation protection
v2lglegplo: Lower chamber legislates in prac...
+ v2smpolcap: Political parties cyber security...
v2exdfvths: HOS veto power in practice
+ v2smlawpr: Defamation protection
v2jupurge: Judicial purges
+ v2lglegplo: Lower chamber legislates in prac...
v2lgoppart: Legislature opposition parties
+ v2smlawpr: Defamation protection
0
40
80
120
Bundle weight (sum of leaf Cover)

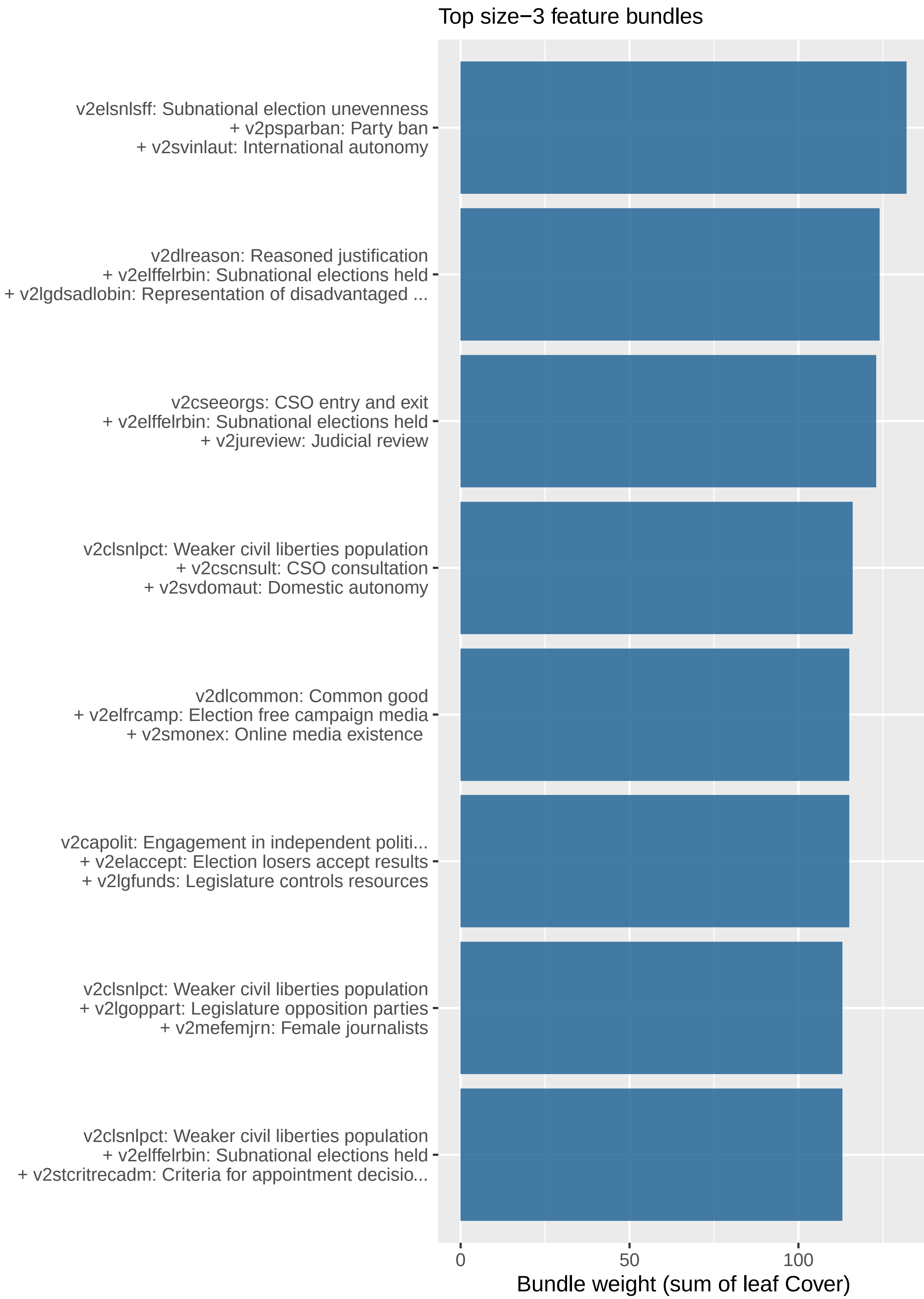

Top size−3 feature bundles
v2castate: Engagement in state−administered...
+ v2excrptps: Public sector corrupt exchanges
+ v2mefemjrn: Female journalists
v2clacfree: Freedom of academic and cultural...
+ v2exdfdshs: HOS dissolution in practice
+ v2jupack: Court packing
v2capolit: Engagement in independent politi...
+ v2lgqstexp: Legislature questions officials ...
+ v2pscnslnl: Candidate selection−national/local
v2elaccept: Election losers accept results
+ v2smprivex: Privacy protection by law exists
+ v2svinlaut: International autonomy
v2caassemb: Freedom of peaceful assembly
+ v2elsnlsff: Subnational election unevenness
+ v2exdfpphs: HOS proposes legislation in prac...
v2excrptps: Public sector corrupt exchanges
+ v2mefemjrn: Female journalists
+ v2smpolcap: Political parties cyber security...
v2psprlnks: Party linkages
+ v2smpolcap: Political parties cyber security...
+ v2smprivex: Privacy protection by law exists
v2clrelig: Freedom of religion
+ v2excrptps: Public sector corrupt exchanges
+ v2lgsrvlo: Lower chamber members serve in g...
0
50
100
150
Bundle weight (sum of leaf Cover)

Infant mortality (log)

N = 170 / best depth = 5 / OOS R^2 = 0.71 / nrounds = 967


v2pehealth: Health equality
v2elvotbuy: Election vote buying
v2peedueq: Educational equality
v2excrptps: Public sector corrupt exchanges
v2smonex: Online media existence
v2elembcap: EMB capacity
v2cacamps: Political polarization
v2smpolcap: Political parties cyber security capa...
v2smgovcapsec: Government cyber security capacity
v2elaccept: Election losers accept results
v2psplats: Distinct party platforms
v2regoppgroupssize: Regime opposition groups size
v2exthftps: Public sector theft
v2elfrfair: Election free and fair
v2smgovsmalt: Government social media alternatives
−0.6
−0.3
0.0
0.3
SHAP value
Feature value
High
Low

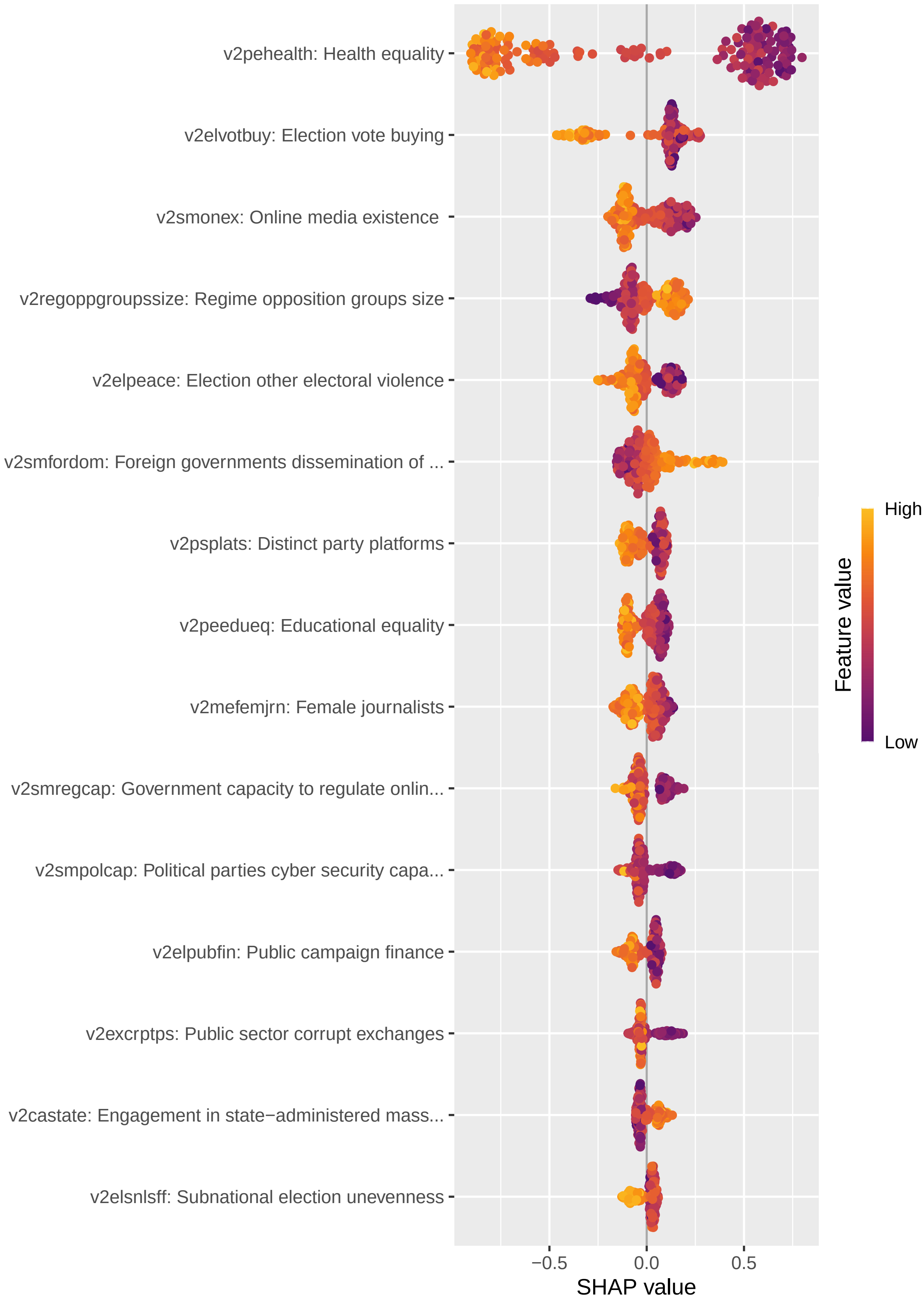

Top size−2 feature bundles
v2ellocpwr: Local offices relative power
+ v2smonper: Online media perspectives
v2elirreg: Election other voting irregulari...
+ v2lgstafflo: Lower chamber staff
v2clsnlpct: Weaker civil liberties population
+ v2lgoppart: Legislature opposition parties
v2clsnlpct: Weaker civil liberties population
+ v2elsnlsff: Subnational election unevenness
v2elsnlsff: Subnational election unevenness
+ v2smpardom: Party dissemination of false inf...
v2elffelrbin: Subnational elections held
+ v2smpolcap: Political parties cyber security...
v2elffelrbin: Subnational elections held
+ v2mefemjrn: Female journalists
v2exdfcbhs_rec: HOS appoints cabinet in practice
+ v2exdfdmhs: HOS dismisses ministers in practice
0
50
100
150
Bundle weight (sum of leaf Cover)

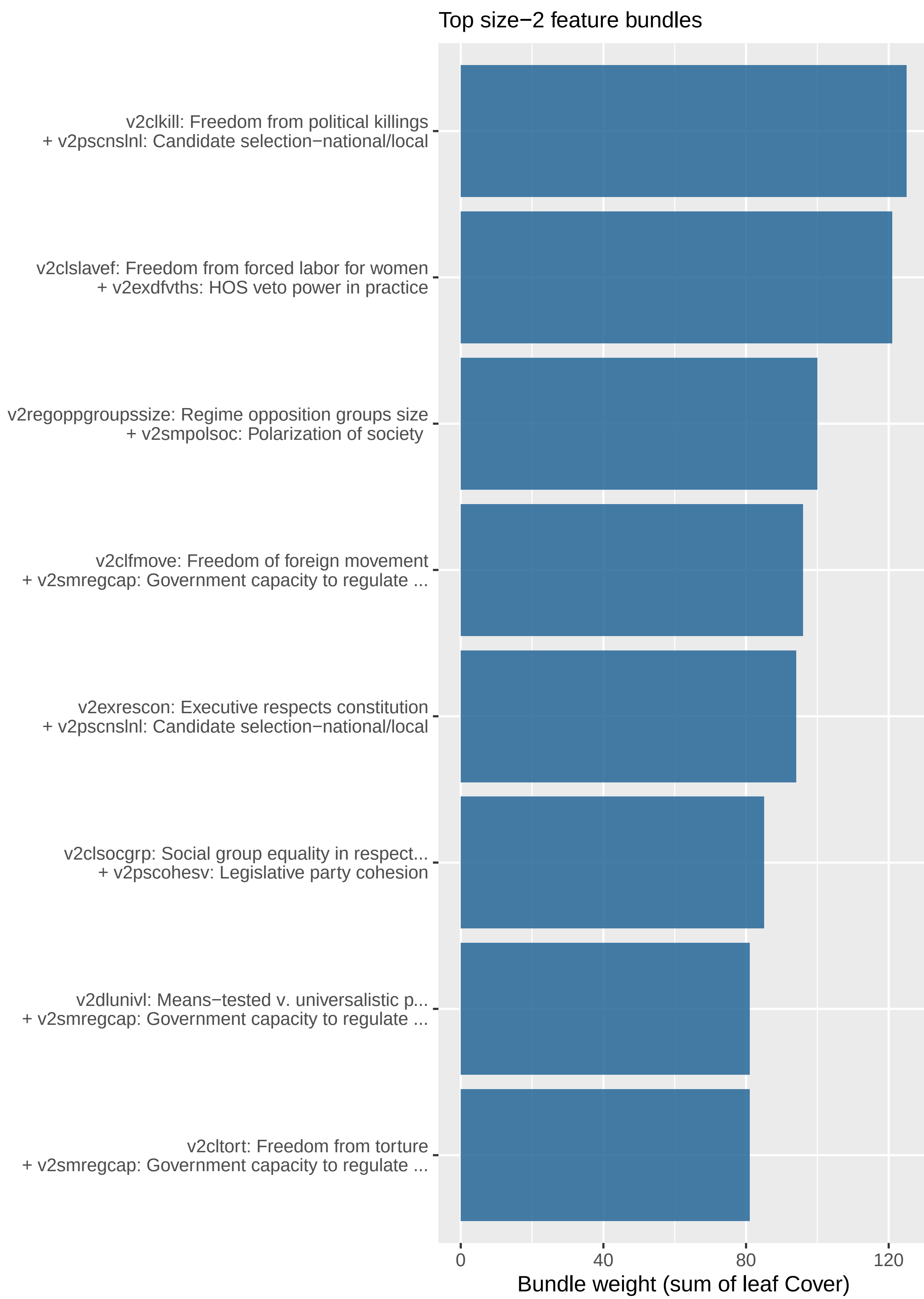

Top size−3 feature bundles
v2elsnlsff: Subnational election unevenness
+ v2psparban: Party ban
+ v2svinlaut: International autonomy
v2dlreason: Reasoned justification
+ v2elffelrbin: Subnational elections held
+ v2lgdsadlobin: Representation of disadvantaged ...
v2cseeorgs: CSO entry and exit
+ v2elffelrbin: Subnational elections held
+ v2jureview: Judicial review
v2clsnlpct: Weaker civil liberties population
+ v2cscnsult: CSO consultation
+ v2svdomaut: Domestic autonomy
v2dlcommon: Common good
+ v2elfrcamp: Election free campaign media
+ v2smonex: Online media existence
v2capolit: Engagement in independent politi...
+ v2elaccept: Election losers accept results
+ v2lgfunds: Legislature controls resources
v2clsnlpct: Weaker civil liberties population
+ v2lgoppart: Legislature opposition parties
+ v2mefemjrn: Female journalists
v2clsnlpct: Weaker civil liberties population
+ v2elffelrbin: Subnational elections held
+ v2stcritrecadm: Criteria for appointment decisio...
0
50
100
Bundle weight (sum of leaf Cover)


Top size−4 feature bundles
v2clsnlpct: Weaker civil liberties population
+ v2dlcommon: Common good
+ v2elffelrbin: Subnational elections held
+ v2svdomaut: Domestic autonomy
v2clsnlpct: Weaker civil liberties population
+ v2elffelrbin: Subnational elections held
+ v2lgoppart: Legislature opposition parties
+ v2meharjrn: Harassment of journalists
v2caassemb: Freedom of peaceful assembly
+ v2clrgunev: Subnational civil liberties unev...
+ v2csrlgcon: Religious organization consultation
+ v2elaccept: Election losers accept results
v2caconmob: Mass mobilization concentration
+ v2dlcommon: Common good
+ v2regoppgroupssize: Regime opposition groups size
+ v2smgovsmmon: Government social media monitoring
v2csrlgcon: Religious organization consultation
+ v2elsnlsff: Subnational election unevenness
+ v2exdfdmhs: HOS dismisses ministers in practice
+ v2jupurge: Judicial purges
v2clsocgrp: Social group equality in respect...
+ v2csreprss: CSO repression
+ v2csrlgcon: Religious organization consultation
+ v2smpolcap: Political parties cyber security...
v2clrgunev: Subnational civil liberties unev...
+ v2mefemjrn: Female journalists
+ v2smfordom: Foreign governments disseminatio...
+ v2smregcon: Internet legal regulation content
v2clrelig: Freedom of religion
+ v2elffelrbin: Subnational elections held
+ v2smpolcap: Political parties cyber security...
+ v2svdomaut: Domestic autonomy
0
50
100
Bundle weight (sum of leaf Cover)

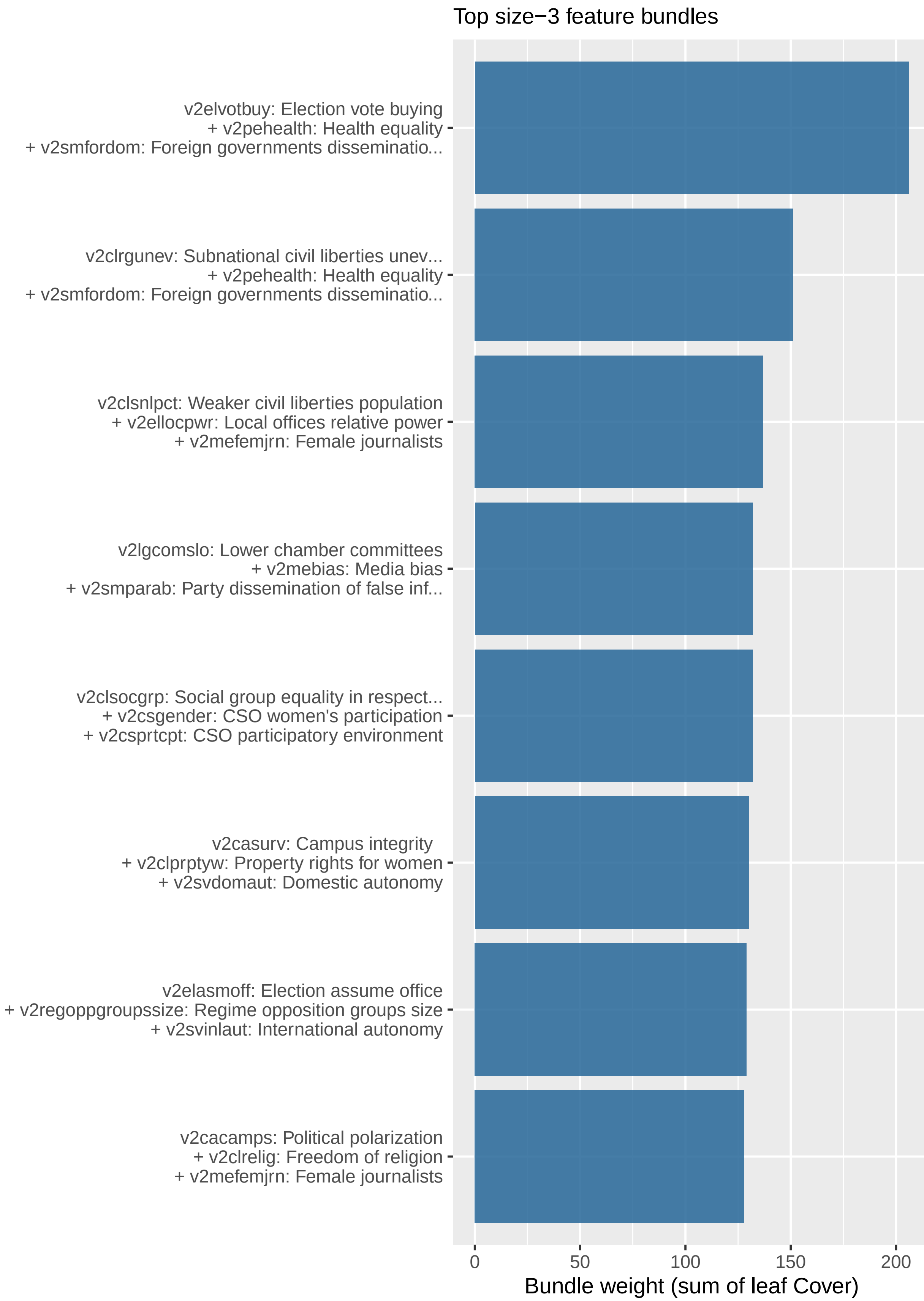
Maternal mortality (log)

N = 170 / best depth = 3 / OOS R^2 = 0.684 / nrounds = 295


v2pehealth: Health equality
v2elvotbuy: Election vote buying
v2smonex: Online media existence
v2regoppgroupssize: Regime opposition groups size
v2elpeace: Election other electoral violence
v2smfordom: Foreign governments dissemination of ...
v2psplats: Distinct party platforms
v2peedueq: Educational equality
v2mefemjrn: Female journalists
v2smregcap: Government capacity to regulate onlin...
v2smpolcap: Political parties cyber security capa...
v2elpubfin: Public campaign finance
v2excrptps: Public sector corrupt exchanges
v2castate: Engagement in state−administered mass...
v2elsnlsff: Subnational election unevenness
−0.5
0.0
0.5
SHAP value
Feature value
High
Low



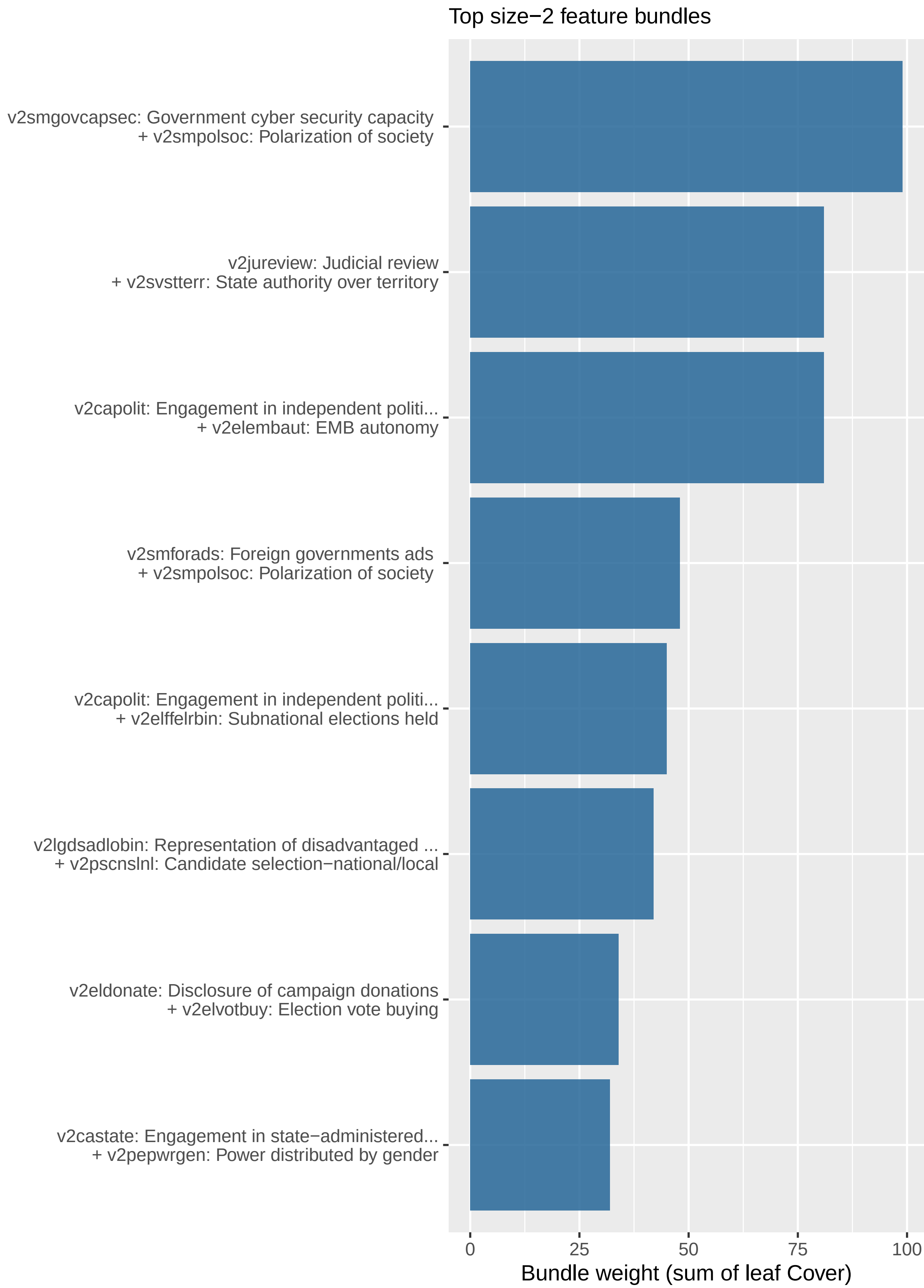
Top size−2 feature bundles
v2clkill: Freedom from political killings
+ v2pscnslnl: Candidate selection−national/local
v2clslavef: Freedom from forced labor for women
+ v2exdfvths: HOS veto power in practice
v2regoppgroupssize: Regime opposition groups size
+ v2smpolsoc: Polarization of society
v2clfmove: Freedom of foreign movement
+ v2smregcap: Government capacity to regulate ...
v2exrescon: Executive respects constitution
+ v2pscnslnl: Candidate selection−national/local
v2clsocgrp: Social group equality in respect...
+ v2pscohesv: Legislative party cohesion
v2dlunivl: Means−tested v. universalistic p...
+ v2smregcap: Government capacity to regulate ...
v2cltort: Freedom from torture
+ v2smregcap: Government capacity to regulate ...
0
40
80
120
Bundle weight (sum of leaf Cover)

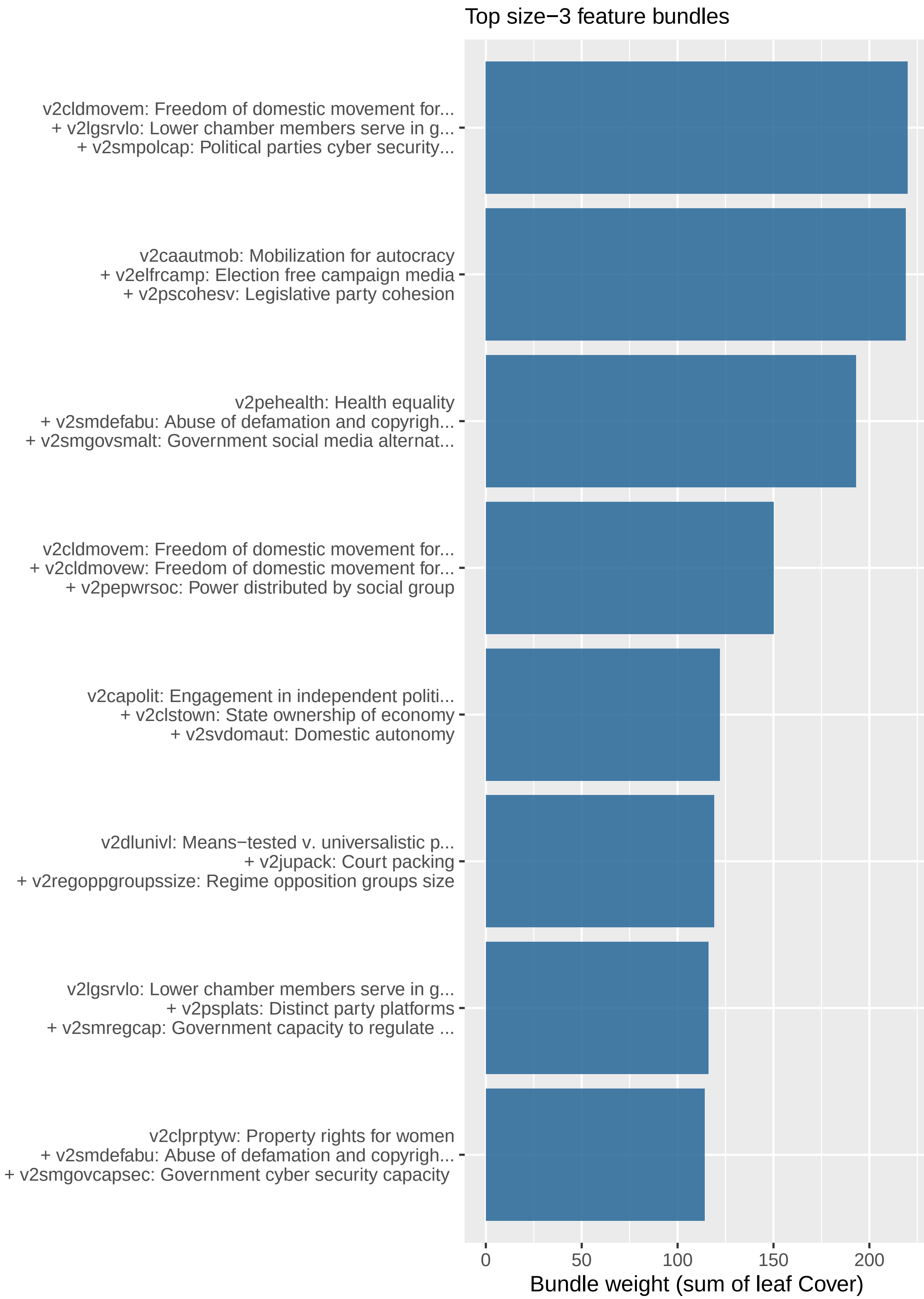

Top size−3 feature bundles
v2elvotbuy: Election vote buying
+ v2pehealth: Health equality
+ v2smfordom: Foreign governments disseminatio...
v2clrgunev: Subnational civil liberties unev...
+ v2pehealth: Health equality
+ v2smfordom: Foreign governments disseminatio...
v2clsnlpct: Weaker civil liberties population
+ v2ellocpwr: Local offices relative power
+ v2mefemjrn: Female journalists
v2lgcomslo: Lower chamber committees
+ v2mebias: Media bias
+ v2smparab: Party dissemination of false inf...
v2clsocgrp: Social group equality in respect...
+ v2csgender: CSO women's participation
+ v2csprtcpt: CSO participatory environment
v2casurv: Campus integrity
+ v2clprptyw: Property rights for women
+ v2svdomaut: Domestic autonomy
v2elasmoff: Election assume office
+ v2regoppgroupssize: Regime opposition groups size
+ v2svinlaut: International autonomy
v2cacamps: Political polarization
+ v2clrelig: Freedom of religion
+ v2mefemjrn: Female journalists
0
50
100
150
200
Bundle weight (sum of leaf Cover)

Human Capital Index

N = 156 / best depth = 3 / OOS R^2 = 0.771 / nrounds = 253

v2peedueq: Educational equality
v2pehealth: Health equality
v2elvotbuy: Election vote buying
v2smonex: Online media existence
v2smgovcapsec: Government cyber security capacity
v2elembcap: EMB capacity
v2stfisccap: State fiscal source of revenue
v2smdefabu: Abuse of defamation and copyright law...
v2castate: Engagement in state−administered mass...
v2psplats: Distinct party platforms
v2clprptyw: Property rights for women
v2stcritapparm: Criteria for appointment decisions in...
v2exthftps: Public sector theft
v2mefemjrn: Female journalists
v2jucorrdc: Judicial corruption decision
−0.025
0.000
0.025
SHAP value
Feature value
High
Low

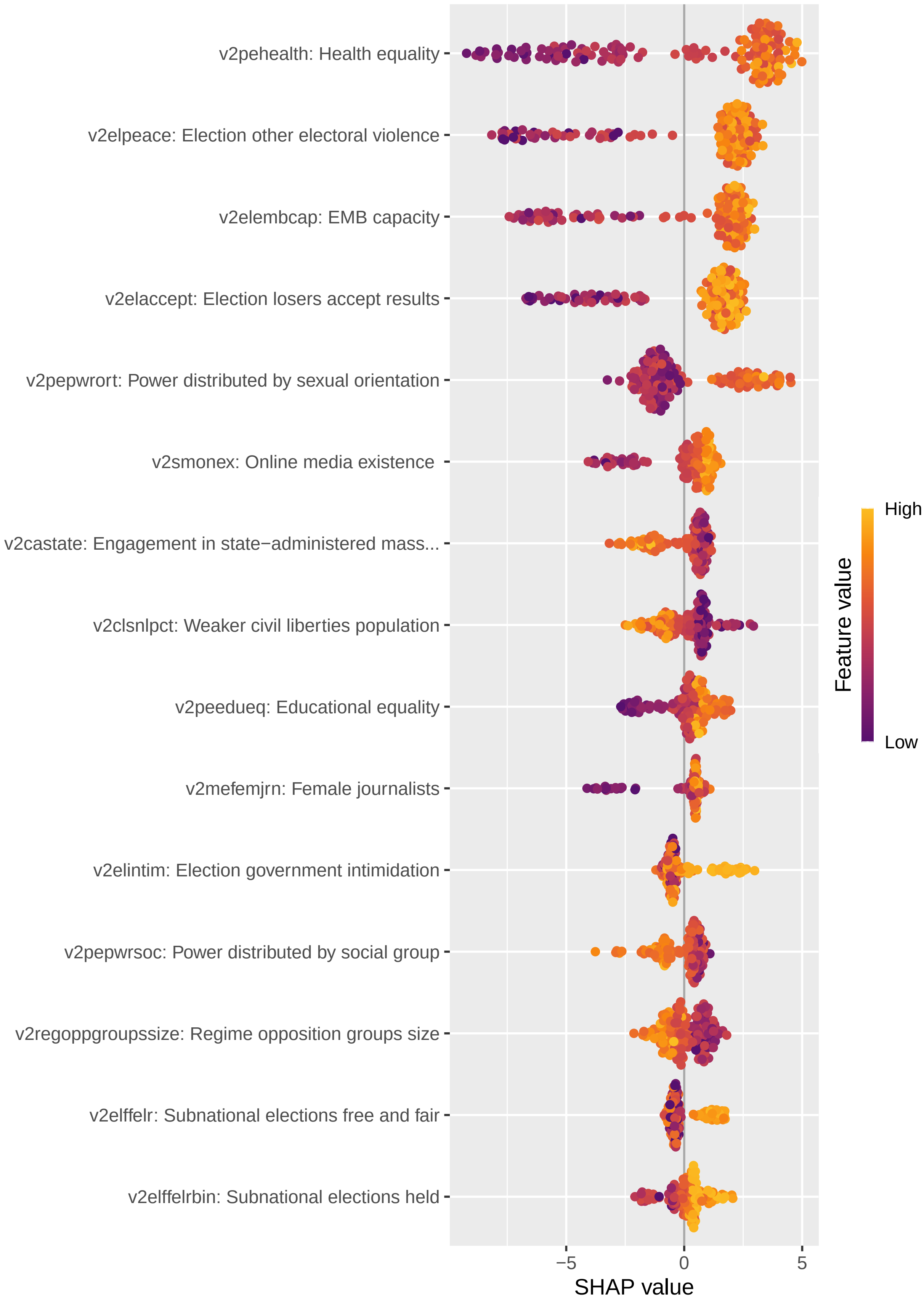

Top size−2 feature bundles
v2smgovcapsec: Government cyber security capacity
+ v2smpolsoc: Polarization of society
v2jureview: Judicial review
+ v2svstterr: State authority over territory
v2capolit: Engagement in independent politi...
+ v2elembaut: EMB autonomy
v2smforads: Foreign governments ads
+ v2smpolsoc: Polarization of society
v2capolit: Engagement in independent politi...
+ v2elffelrbin: Subnational elections held
v2lgdsadlobin: Representation of disadvantaged ...
+ v2pscnslnl: Candidate selection−national/local
v2eldonate: Disclosure of campaign donations
+ v2elvotbuy: Election vote buying
v2castate: Engagement in state−administered...
+ v2pepwrgen: Power distributed by gender
0
25
50
75
100
Bundle weight (sum of leaf Cover)



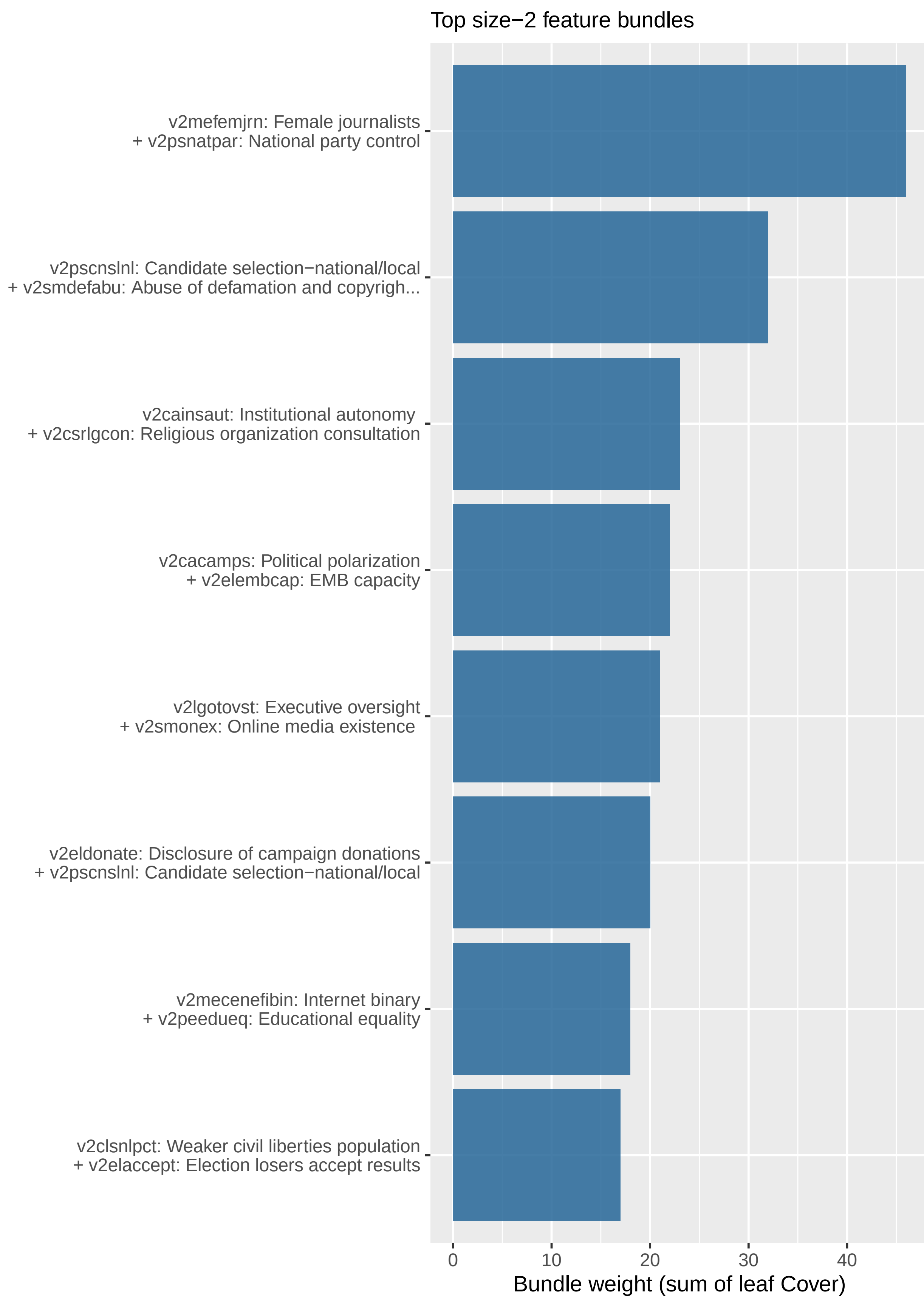
Top size−3 feature bundles
v2cldmovem: Freedom of domestic movement for...
+ v2lgsrvlo: Lower chamber members serve in g...
+ v2smpolcap: Political parties cyber security...
v2caautmob: Mobilization for autocracy
+ v2elfrcamp: Election free campaign media
+ v2pscohesv: Legislative party cohesion
v2pehealth: Health equality
+ v2smdefabu: Abuse of defamation and copyrigh...
+ v2smgovsmalt: Government social media alternat...
v2cldmovem: Freedom of domestic movement for...
+ v2cldmovew: Freedom of domestic movement for...
+ v2pepwrsoc: Power distributed by social group
v2capolit: Engagement in independent politi...
+ v2clstown: State ownership of economy
+ v2svdomaut: Domestic autonomy
v2dlunivl: Means−tested v. universalistic p...
+ v2jupack: Court packing
+ v2regoppgroupssize: Regime opposition groups size
v2lgsrvlo: Lower chamber members serve in g...
+ v2psplats: Distinct party platforms
+ v2smregcap: Government capacity to regulate ...
v2clprptyw: Property rights for women
+ v2smdefabu: Abuse of defamation and copyrigh...
+ v2smgovcapsec: Government cyber security capacity
0
50
100
150
200
Bundle weight (sum of leaf Cover)

Secondary enrollment %

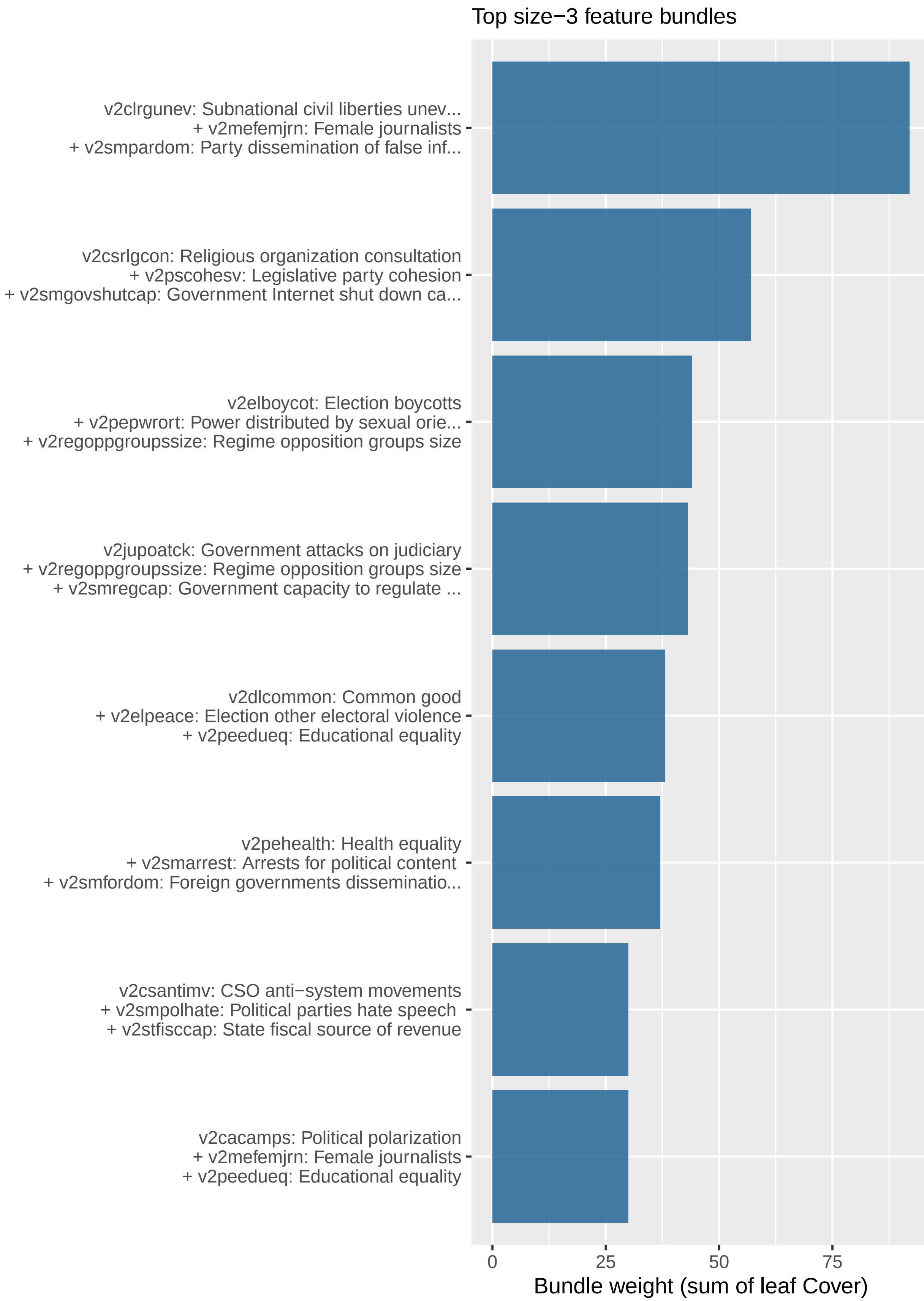
N = 144 / best depth = 4 / OOS R^2 = 0.517 / nrounds = 150



v2pehealth: Health equality
v2elpeace: Election other electoral violence
v2elembcap: EMB capacity
v2elaccept: Election losers accept results
v2pepwrort: Power distributed by sexual orientation
v2smonex: Online media existence
v2castate: Engagement in state−administered mass...
v2clsnlpct: Weaker civil liberties population
v2peedueq: Educational equality
v2mefemjrn: Female journalists
v2elintim: Election government intimidation
v2pepwrsoc: Power distributed by social group
v2regoppgroupssize: Regime opposition groups size
v2elffelr: Subnational elections free and fair
v2elffelrbin: Subnational elections held
−5
0
5
SHAP value
High
Feature value
Low



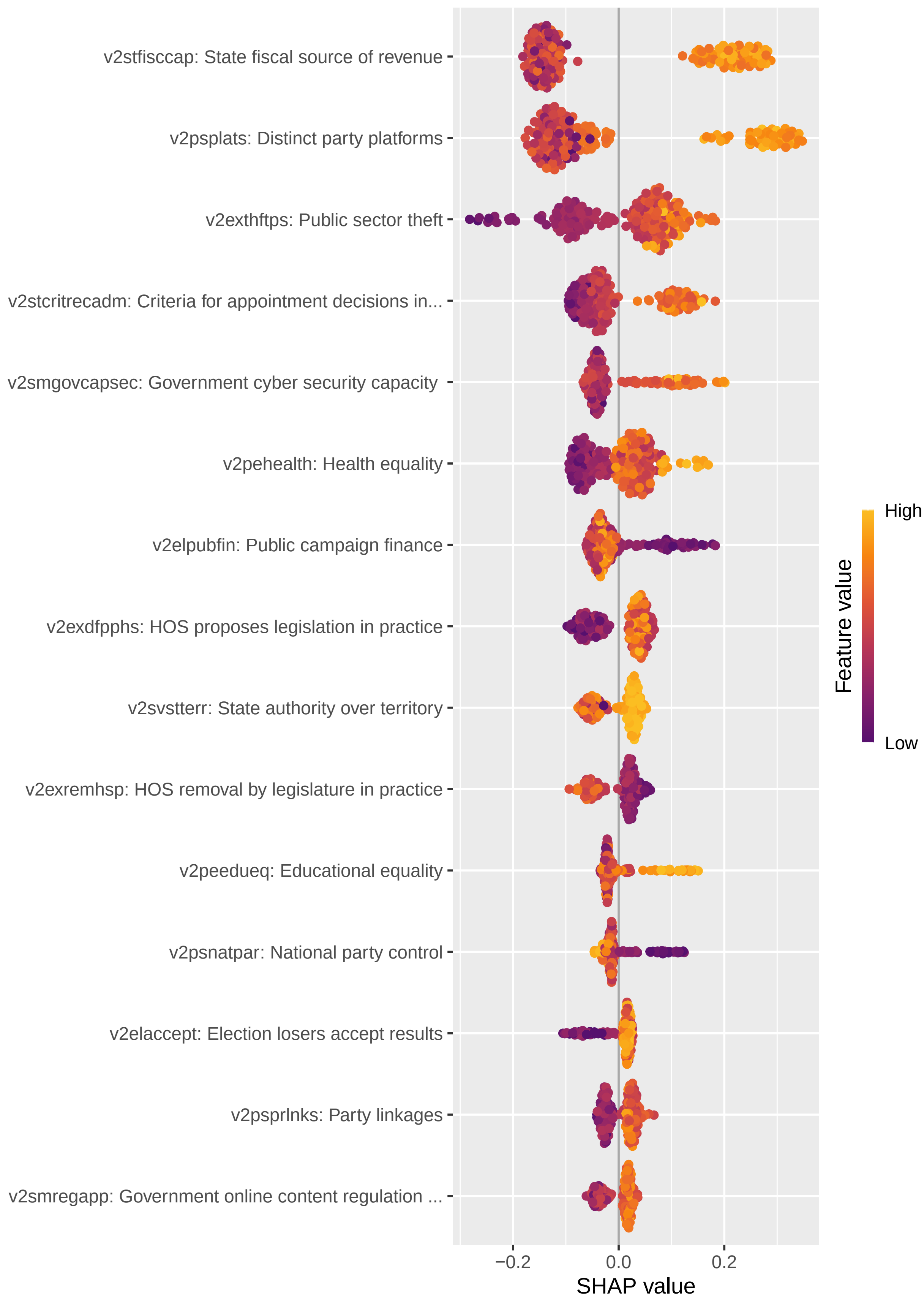
Top size−2 feature bundles
v2mefemjrn: Female journalists
+ v2psnatpar: National party control
v2pscnslnl: Candidate selection−national/local
+ v2smdefabu: Abuse of defamation and copyrigh...
v2cainsaut: Institutional autonomy
+ v2csrlgcon: Religious organization consultation
v2cacamps: Political polarization
+ v2elembcap: EMB capacity
v2lgotovst: Executive oversight
+ v2smonex: Online media existence
v2eldonate: Disclosure of campaign donations
+ v2pscnslnl: Candidate selection−national/local
v2mecenefibin: Internet binary
+ v2peedueq: Educational equality
v2clsnlpct: Weaker civil liberties population
+ v2elaccept: Election losers accept results
0
10
20
30
40
Bundle weight (sum of leaf Cover)


Top size−3 feature bundles
v2clrgunev: Subnational civil liberties unev...
+ v2mefemjrn: Female journalists
+ v2smpardom: Party dissemination of false inf...
v2csrlgcon: Religious organization consultation
+ v2pscohesv: Legislative party cohesion
+ v2smgovshutcap: Government Internet shut down ca...
v2elboycot: Election boycotts
+ v2pepwrort: Power distributed by sexual orie...
+ v2regoppgroupssize: Regime opposition groups size
v2jupoatck: Government attacks on judiciary
+ v2regoppgroupssize: Regime opposition groups size
+ v2smregcap: Government capacity to regulate ...
v2dlcommon: Common good
+ v2elpeace: Election other electoral violence
+ v2peedueq: Educational equality
v2pehealth: Health equality
+ v2smarrest: Arrests for political content
+ v2smfordom: Foreign governments disseminatio...
v2csantimv: CSO anti−system movements
+ v2smpolhate: Political parties hate speech
+ v2stfisccap: State fiscal source of revenue
v2cacamps: Political polarization
+ v2mefemjrn: Female journalists
+ v2peedueq: Educational equality
0
25
50
75
Bundle weight (sum of leaf Cover)

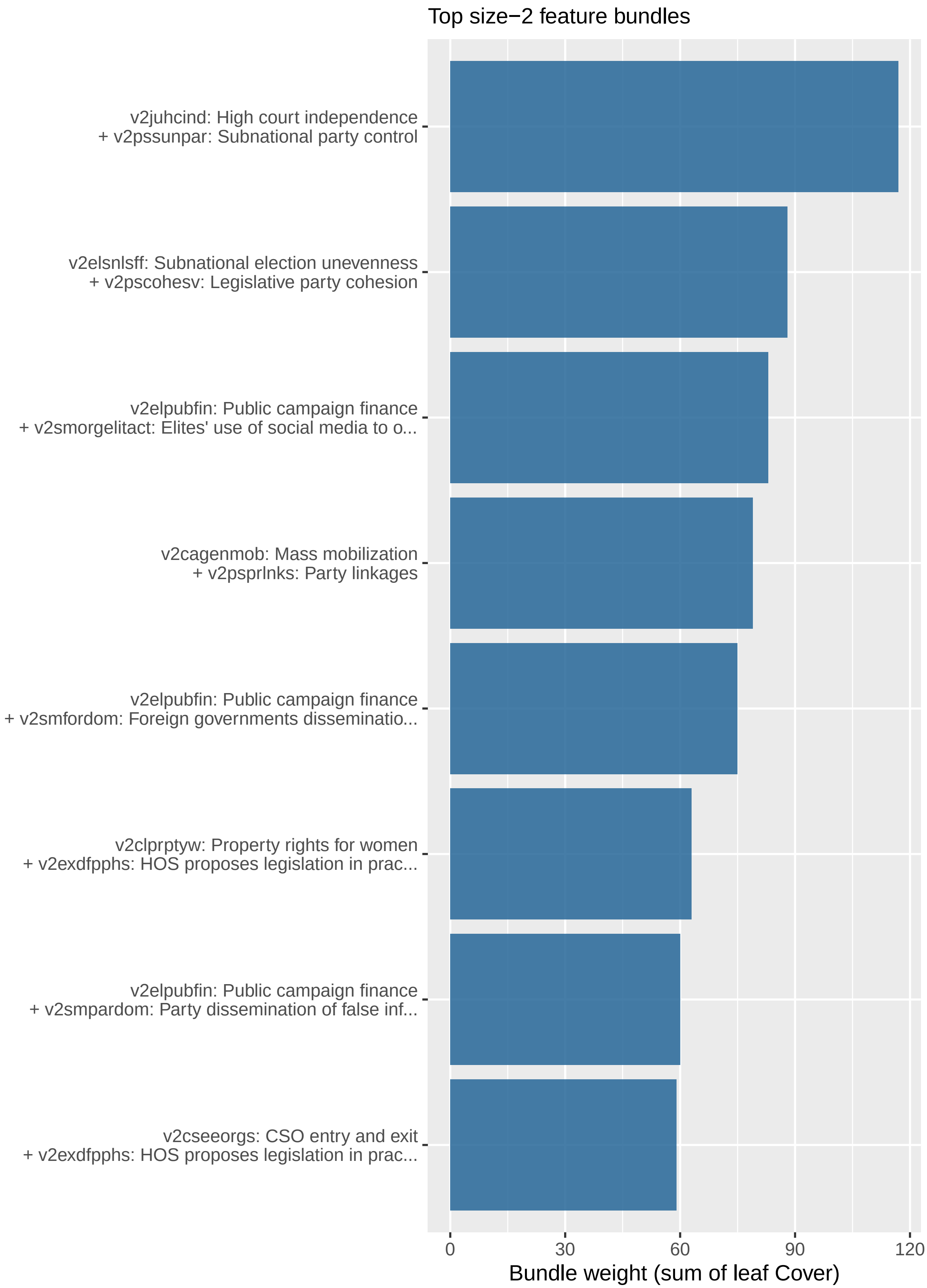
Top size−4 feature bundles
v2catrauni: Engagement in independent trade ...
+ v2clrgunev: Subnational civil liberties unev...
+ v2pepwrort: Power distributed by sexual orie...
+ v2pscnslnl: Candidate selection−national/local
v2clprptyw: Property rights for women
+ v2csrlgcon: Religious organization consultation
+ v2smgovshutcap: Government Internet shut down ca...
+ v2smregapp: Government online content regula...
v2cainsaut: Institutional autonomy
+ v2cldmovew: Freedom of domestic movement for...
+ v2pepwrsoc: Power distributed by social group
+ v2pssunpar: Subnational party control
v2csgender: CSO women's participation
+ v2exbribe: Executive bribery and corrupt ex...
+ v2exremhsp: HOS removal by legislature in pr...
+ v2pscnslnl: Candidate selection−national/local
v2cafexch: Freedom of academic exchange and...
+ v2elmulpar: Elections multiparty
+ v2pepwrgen: Power distributed by gender
+ v2pscnslnl: Candidate selection−national/local
v2clacjstm: Access to justice for men
+ v2cltort: Freedom from torture
+ v2dlcountr: Respect counterarguments
+ v2exremhsp: HOS removal by legislature in pr...
v2capolit: Engagement in independent politi...
+ v2elboycot: Election boycotts
+ v2mefemjrn: Female journalists
+ v2smfordom: Foreign governments disseminatio...
v2cafexch: Freedom of academic exchange and...
+ v2csgender: CSO women's participation
+ v2csrlgcon: Religious organization consultation
+ v2elffelrbin: Subnational elections held
0
25
50
75
100
Bundle weight (sum of leaf Cover)

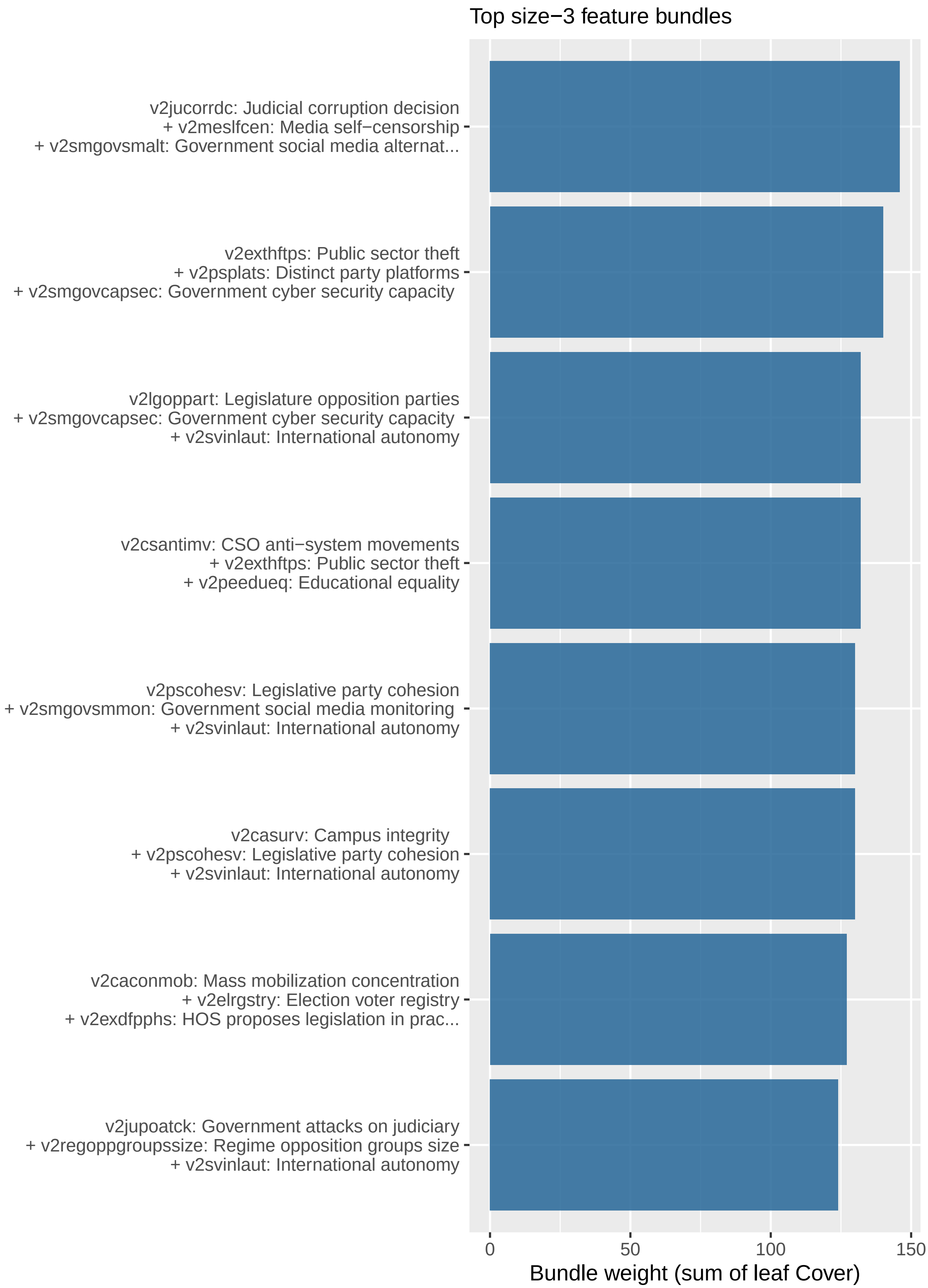
Economic Complexity (ECI)

N = 172 / best depth = 3 / OOS R^2 = 0.45 / nrounds = 136



v2stfisccap: State fiscal source of revenue
v2psplats: Distinct party platforms
v2exthftps: Public sector theft
v2stcritrecadm: Criteria for appointment decisions in...
v2smgovcapsec: Government cyber security capacity
v2pehealth: Health equality
v2elpubfin: Public campaign finance
v2exdfpphs: HOS proposes legislation in practice
v2svstterr: State authority over territory
v2exremhsp: HOS removal by legislature in practice
v2peedueq: Educational equality
v2psnatpar: National party control
v2elaccept: Election losers accept results
v2psprlnks: Party linkages
v2smregapp: Government online content regulation ...
−0.2
0.0
0.2
SHAP value
Feature value
High
Low

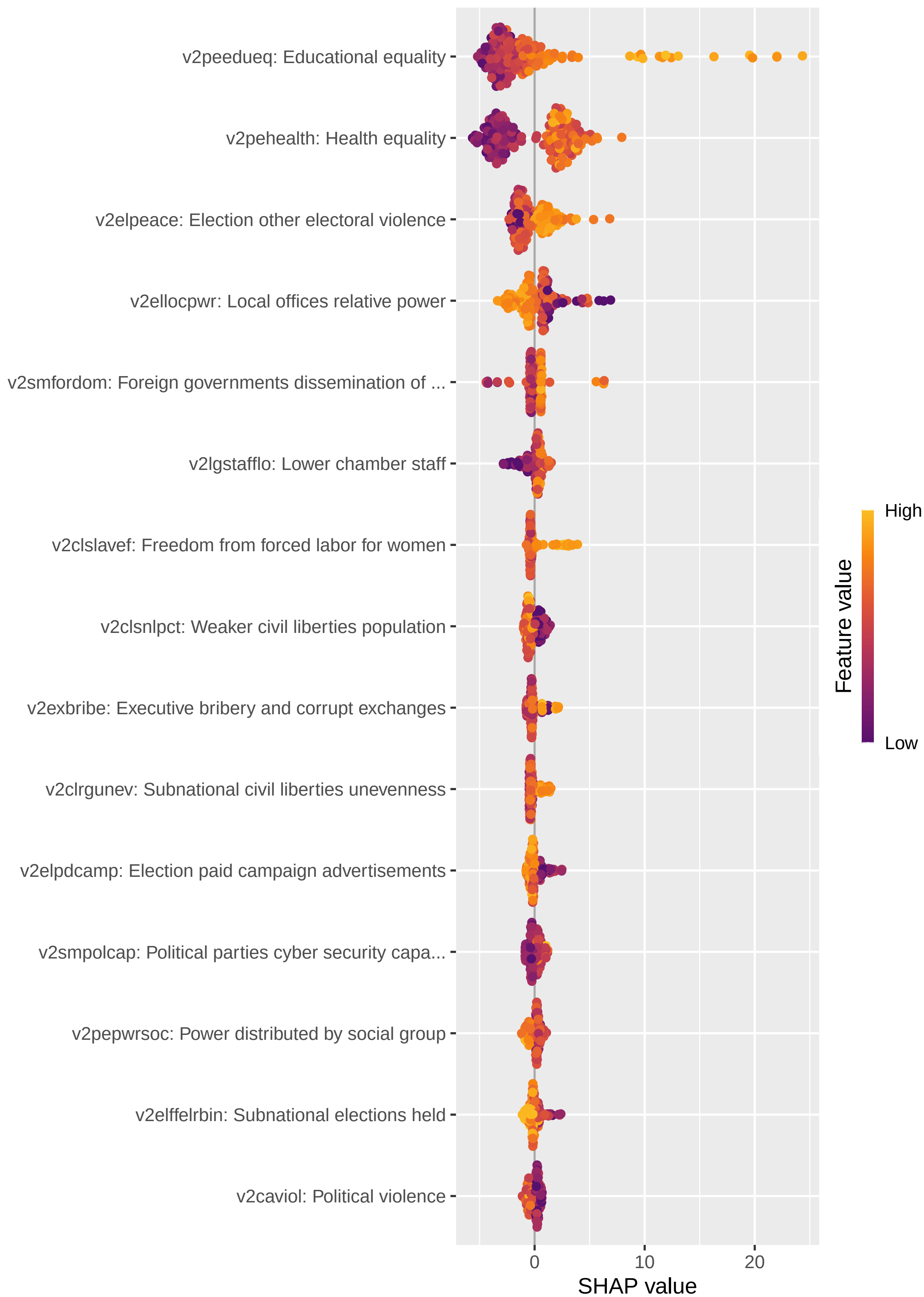

Top size−2 feature bundles
v2juhcind: High court independence
+ v2pssunpar: Subnational party control
v2elsnlsff: Subnational election unevenness
+ v2pscohesv: Legislative party cohesion
v2elpubfin: Public campaign finance
+ v2smorgelitact: Elites' use of social media to o...
v2cagenmob: Mass mobilization
+ v2psprlnks: Party linkages
v2elpubfin: Public campaign finance
+ v2smfordom: Foreign governments disseminatio...
v2clprptyw: Property rights for women
+ v2exdfpphs: HOS proposes legislation in prac...
v2elpubfin: Public campaign finance
+ v2smpardom: Party dissemination of false inf...
v2cseeorgs: CSO entry and exit
+ v2exdfpphs: HOS proposes legislation in prac...
0
30
60
90
120
Bundle weight (sum of leaf Cover)

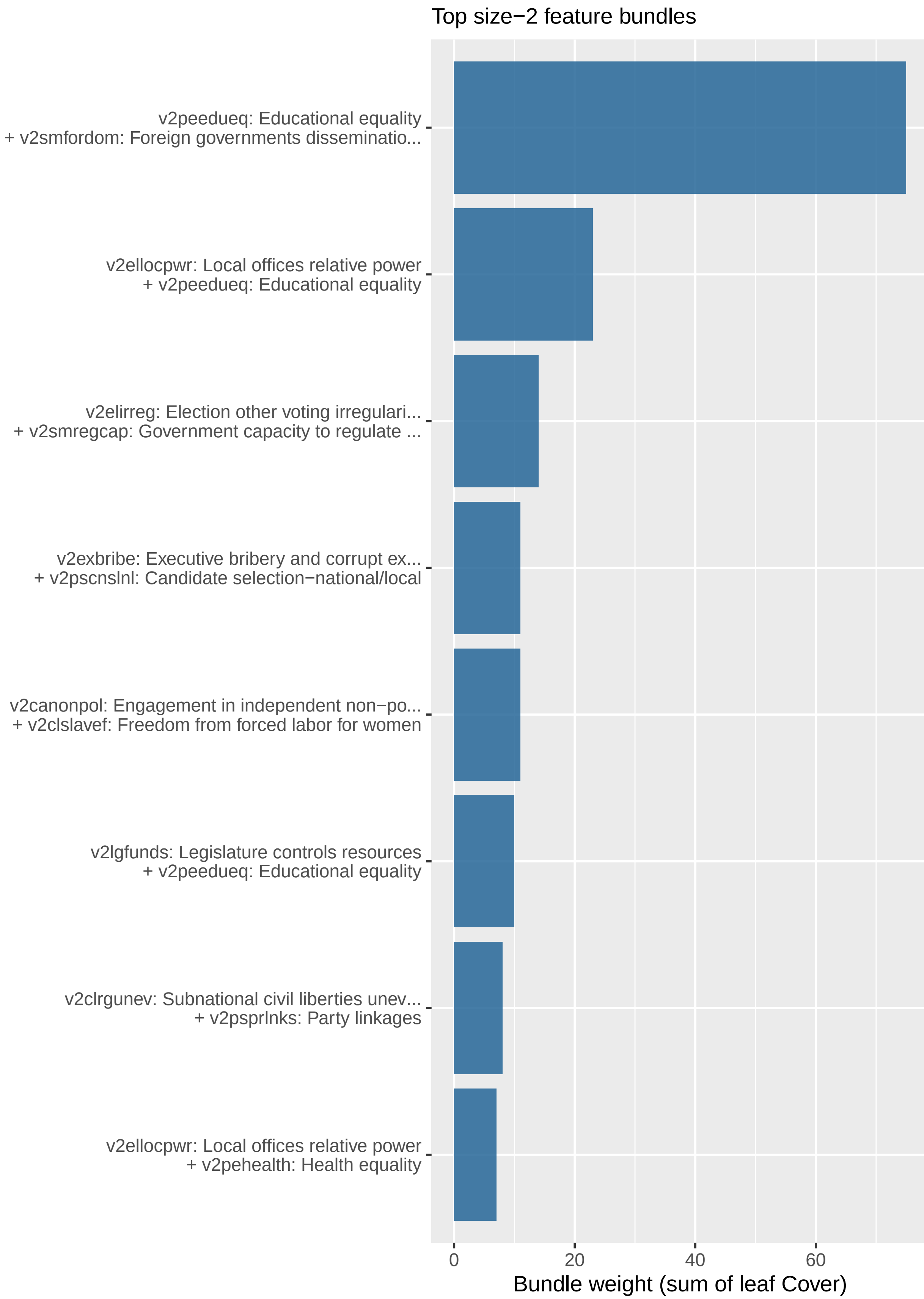


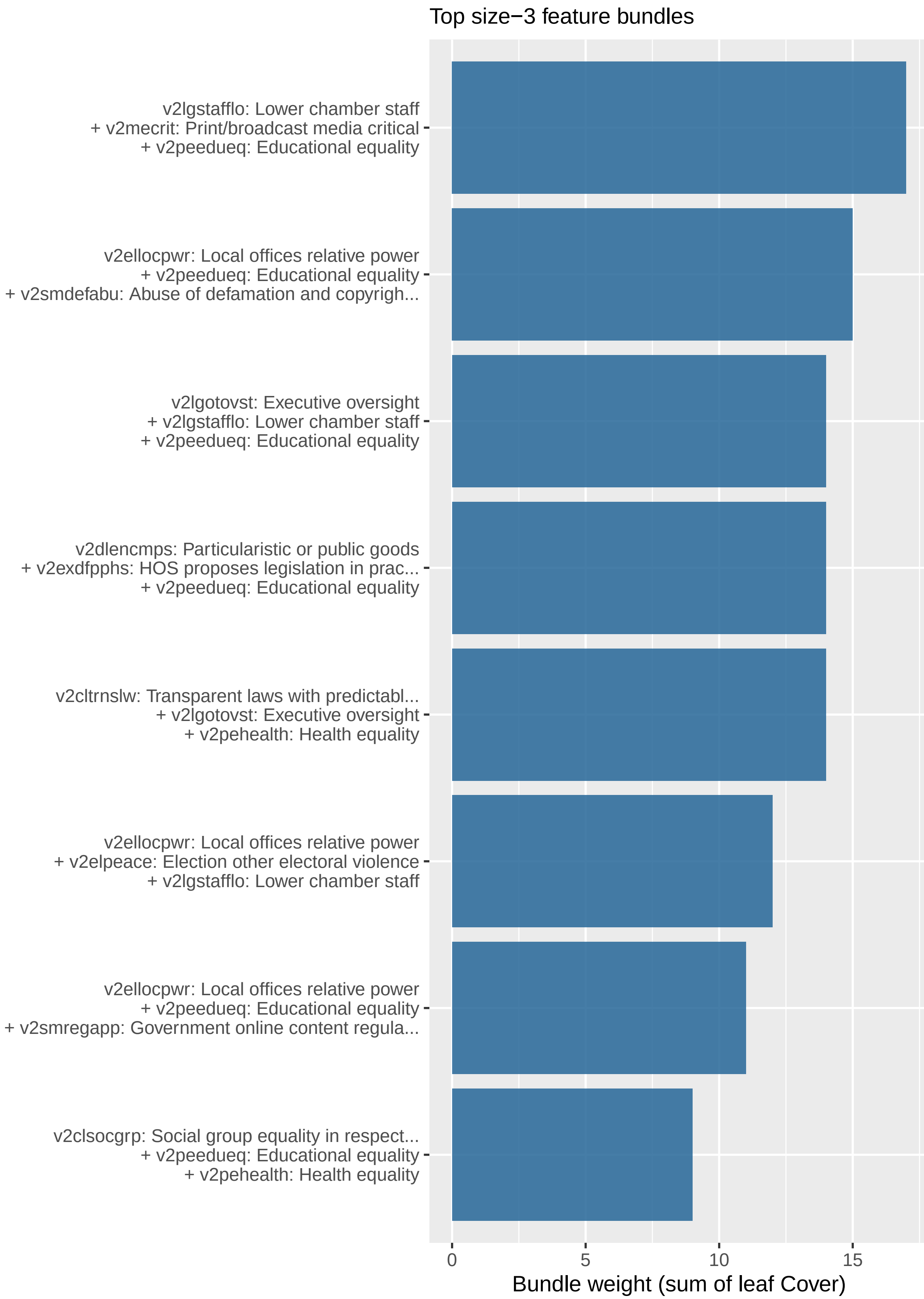
Top size−3 feature bundles
v2jucorrdc: Judicial corruption decision
+ v2meslfcen: Media self−censorship
+ v2smgovsmalt: Government social media alternat...
v2exthftps: Public sector theft
+ v2psplats: Distinct party platforms
+ v2smgovcapsec: Government cyber security capacity
v2lgoppart: Legislature opposition parties
+ v2smgovcapsec: Government cyber security capacity
+ v2svinlaut: International autonomy
v2csantimv: CSO anti−system movements
+ v2exthftps: Public sector theft
+ v2peedueq: Educational equality
v2pscohesv: Legislative party cohesion
+ v2smgovsmmon: Government social media monitoring
+ v2svinlaut: International autonomy
v2casurv: Campus integrity
+ v2pscohesv: Legislative party cohesion
+ v2svinlaut: International autonomy
v2caconmob: Mass mobilization concentration
+ v2elrgstry: Election voter registry
+ v2exdfpphs: HOS proposes legislation in prac...
v2jupoatck: Government attacks on judiciary
+ v2regoppgroupssize: Regime opposition groups size
+ v2svinlaut: International autonomy
0
50
100
150
Bundle weight (sum of leaf Cover)

Exports % GDP

N = 155 / best depth = 6 / OOS R^2 = 0.155 / nrounds = 22


v2peedueq: Educational equality
v2pehealth: Health equality
v2elpeace: Election other electoral violence
v2ellocpwr: Local offices relative power
v2smfordom: Foreign governments dissemination of ...
v2lgstafflo: Lower chamber staff
v2clslavef: Freedom from forced labor for women
v2clsnlpct: Weaker civil liberties population
v2exbribe: Executive bribery and corrupt exchanges
v2clrgunev: Subnational civil liberties unevenness
v2elpdcamp: Election paid campaign advertisements
v2smpolcap: Political parties cyber security capa...
v2pepwrsoc: Power distributed by social group
v2elffelrbin: Subnational elections held
v2caviol: Political violence
0
10
20
SHAP value
Feature value
High
Low


Top size−2 feature bundles
v2peedueq: Educational equality
+ v2smfordom: Foreign governments disseminatio...
v2ellocpwr: Local offices relative power
+ v2peedueq: Educational equality
v2elirreg: Election other voting irregulari...
+ v2smregcap: Government capacity to regulate ...
v2exbribe: Executive bribery and corrupt ex...
+ v2pscnslnl: Candidate selection−national/local
v2canonpol: Engagement in independent non−po...
+ v2clslavef: Freedom from forced labor for women
v2lgfunds: Legislature controls resources
+ v2peedueq: Educational equality
v2clrgunev: Subnational civil liberties unev...
+ v2psprlnks: Party linkages
v2ellocpwr: Local offices relative power
+ v2pehealth: Health equality
0
20
40
60
Bundle weight (sum of leaf Cover)


Top size−3 feature bundles
v2lgstafflo: Lower chamber staff
+ v2mecrit: Print/broadcast media critical
+ v2peedueq: Educational equality
v2ellocpwr: Local offices relative power
+ v2peedueq: Educational equality
+ v2smdefabu: Abuse of defamation and copyrigh...
v2lgotovst: Executive oversight
+ v2lgstafflo: Lower chamber staff
+ v2peedueq: Educational equality
v2dlencmps: Particularistic or public goods
+ v2exdfpphs: HOS proposes legislation in prac...
+ v2peedueq: Educational equality
v2cltrnslw: Transparent laws with predictabl...
+ v2lgotovst: Executive oversight
+ v2pehealth: Health equality
v2ellocpwr: Local offices relative power
+ v2elpeace: Election other electoral violence
+ v2lgstafflo: Lower chamber staff
v2ellocpwr: Local offices relative power
+ v2peedueq: Educational equality
+ v2smregapp: Government online content regula...
v2clsocgrp: Social group equality in respect...
+ v2peedueq: Educational equality
+ v2pehealth: Health equality
0
5
10
15
Bundle weight (sum of leaf Cover)


Top size−4 feature bundles
v2caviol: Political violence
+ v2clslavef: Freedom from forced labor for women
+ v2lgcrrpt: Legislature corrupt activities
+ v2pepwrort: Power distributed by sexual orie...
v2excrptps: Public sector corrupt exchanges
+ v2pehealth: Health equality
+ v2pscnslnl: Candidate selection−national/local
+ v2smpolhate: Political parties hate speech
v2merange: Print/broadcast media perspectives
+ v2pehealth: Health equality
+ v2pepwrses: Power distributed by socioeconom...
+ v2pscnslnl: Candidate selection−national/local
v2elpdcamp: Election paid campaign advertise...
+ v2lgstafflo: Lower chamber staff
+ v2peedueq: Educational equality
+ v2pehealth: Health equality
v2elpaidig: Election paid interest group media
+ v2lgstafflo: Lower chamber staff
+ v2peedueq: Educational equality
+ v2smpolcap: Political parties cyber security...
v2ellocpwr: Local offices relative power
+ v2exdfcbhs: HOS appoints cabinet in practice
+ v2exdfdmhs: HOS dismisses ministers in practice
+ v2pehealth: Health equality
v2clrgunev: Subnational civil liberties unev...
+ v2elirreg: Election other voting irregulari...
+ v2jupurge: Judicial purges
+ v2psprlnks: Party linkages
v2caviol: Political violence
+ v2elpeace: Election other electoral violence
+ v2pehealth: Health equality
+ v2smregcap: Government capacity to regulate ...
0
10
20
Bundle weight (sum of leaf Cover)

Two-PC summary

The ML model treats each of ~60 V-Dem features as its own knob. But these features are highly collinear: if we run a PCA on the Mcp matrix, two or three principal components will usually capture most of the variation across countries. That suggests a simpler exercise: **project each country into the first two principal components, split the plane into quadrants, and ask what each quadrant means for outcomes.** If the first two PCs explain most of the outcome signal too, the quadrant labels tell a compact story about what bundles of institutional arrangements tend to produce what outcomes.

PCA on Mcp

We center and scale each V-Dem feature before PCA (`prcomp` with `scale. = TRUE`) so that the min-max normalisation earlier in the pipeline doesn't let wider-variance features dominate the rotation.

Table 5: Share of Mcp variance explained by the first 10 principal components.

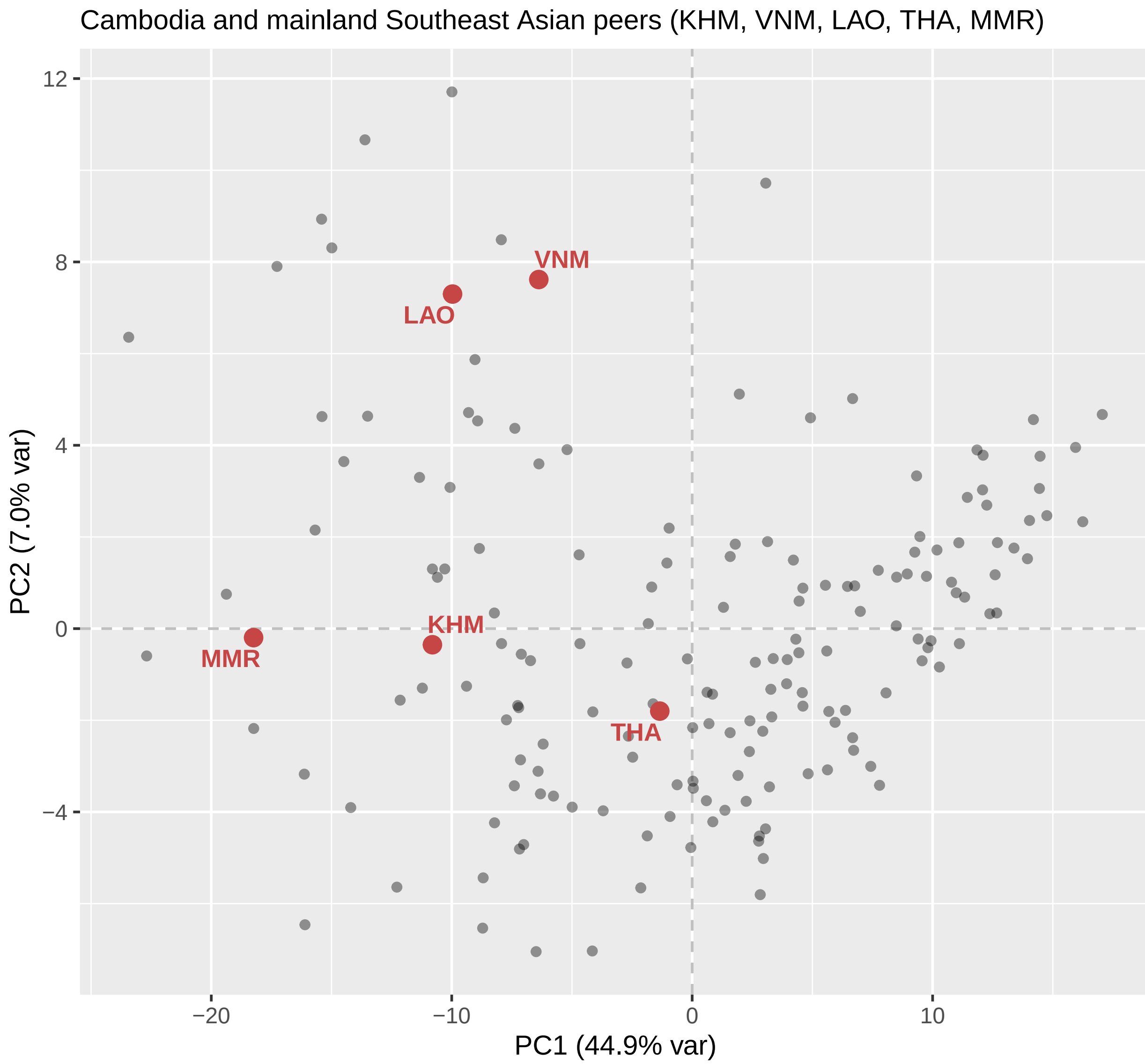
PC	Variance explained	Cumulative
1	0.449	0.449
2	0.070	0.518
3	0.040	0.559
4	0.027	0.586
5	0.025	0.611
6	0.018	0.629
7	0.017	0.647
8	0.015	0.662
9	0.014	0.676
10	0.013	0.688

Table 6: Top 10 V-Dem features by absolute loading on PC1 and PC2. Loadings are signed; sign indicates direction along the axis.

axis	code	description	loading
PC1	v2elembaut	EMB autonomy	0.101
PC1	v2cltort	Freedom from torture	0.099
PC1	v2cldiscw	Freedom of discussion for women	0.098
PC1	v2cldiscm	Freedom of discussion for men	0.098
PC1	v2meharjrn	Harassment of journalists	0.098
PC1	v2smarrest	Arrests for political content	0.097
PC1	v2csreprss	CSO repression	0.097
PC1	v2elfrfair	Election free and fair	0.097
PC1	v2mecenefm	Government censorship effort — Media	0.096
PC1	v2clfmove	Freedom of foreign movement	0.096
PC2	v2csantimv	CSO anti-system movements	-0.181
PC2	v2smpolsoc	Polarization of society	0.172
PC2	v2cademmob	Mobilization for democracy	-0.171
PC2	v2caviol	Political violence	-0.169
PC2	v2smorgviol	Use of social media to organize offline vi…	0.160
PC2	v2cagenmob	Mass mobilization	-0.156
PC2	v2smmefra	Online media fractionalization	0.153

axis	code	description	loading
PC2	v2cacamps	Political polarization	-0.149
PC2	v2pehealth	Health equality	0.145
PC2	v2smorgelitact	Elites’ use of social media to organize of. . .	-0.145

Country scatter



Cambodia's institutional peers

Geography is a guess at peerhood. With the PCA in hand we can ask the data who Cambodia's actual institutional neighbours are — countries with similar V-Dem fingerprints rather than similar coordinates on a map. We compute Euclidean distance from Cambodia to every other country in the first ten PCs (about 69% of total variance) and rank.

Table 7: Cambodia's 15 nearest institutional peers in the top-10 PC subspace (cumulative variance 68.8%). Distance is Euclidean in PC space, which weights each PC by its variance.

rank	iso3	country	distance	PC1	PC2
1	BDI	Burundi	5.41	-11.22	-1.30
2	EGY	Egypt	7.54	-10.07	3.08
3	ETH	Ethiopia	8.04	-7.93	-0.33

rank	iso3	country	distance	PC1	PC2
4	ZWE	Zimbabwe	8.06	-7.21	-1.72
5	COG	Republic of the Congo	8.22	-9.38	-1.26
6	DJI	Djibouti	8.49	-8.85	1.75
7	RUS	Russia	8.51	-11.34	3.30
8	UGA	Uganda	8.53	-7.11	-0.56
9	MRT	Mauritania	8.76	-7.26	-1.68
10	UZB	Uzbekistan	8.78	-9.30	4.71
11	PAK	Pakistan	8.87	-5.77	-3.65
12	KAZ	Kazakhstan	8.93	-6.37	3.59
13	SLV	El Salvador	8.98	-6.20	-2.52
14	IRN	Iran	9.27	-10.59	1.12
15	TUR	Türkiye	9.31	-7.72	-1.99

Cambodia's 8 nearest institutional peers (top−10 PC subspace)

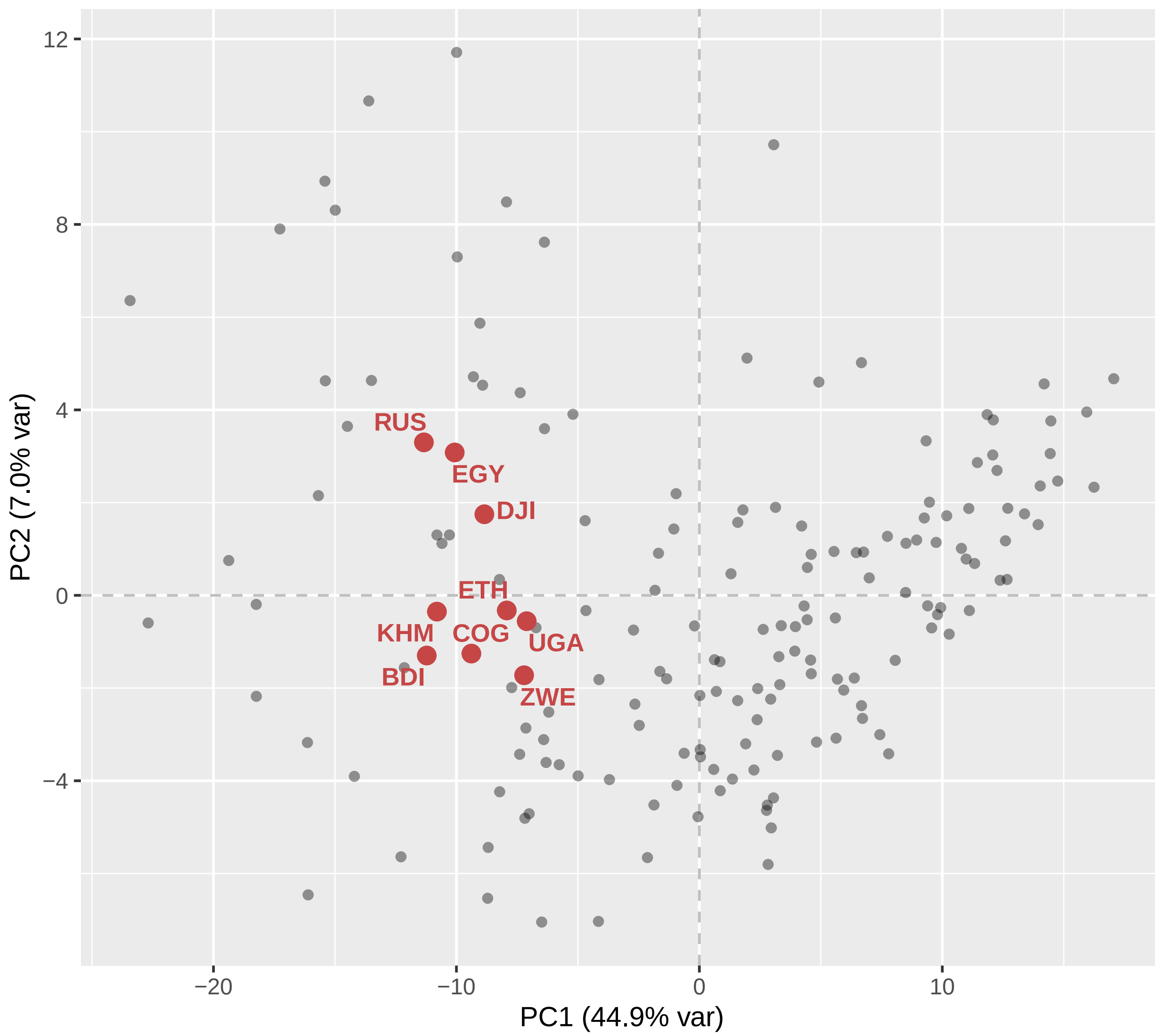


Table 8: Where the geographic neighbours rank in Cambodia's institutional-similarity ordering. The closest is well outside the top 25 — geography and institutions disagree.

iso3	country	rank	distance	PC1	PC2
LAO	Laos	27	10.04	-9.97	7.30
VNM	Vietnam	32	10.20	-6.38	7.62
MMR	Burma/Myanmar	35	10.43	-18.24	-0.20
THA	Thailand	45	11.72	-1.35	-1.80

Cambodia and peers: actual vs predicted outcomes

For each outcome we ask: **given this country's V-Dem fingerprint, what would a model that hasn't seen it expect its outcome to be?** The honest version is **leave-one-out**: for each focus country and each outcome, refit the best-depth model on every *other* country and predict the held-out one. Comparing this prediction to the country's actual outcome surfaces *institutional surprises* — countries that outperform or underperform what their institutional setup would predict.

The residual is rescaled by the model's out-of-sample RMSE (derived from the 5-fold CV R^2 in the depth sweep above), so a z-score of +1 means the country's actual outcome is one OOS-RMSE above what the model predicts.

We look at six focus countries: Cambodia (KHM), its institutional twin Burundi (BDI), and its four mainland Southeast Asian geographic neighbours (VNM, LAO, THA, MMR). Outcomes with `log` transforms (infant and maternal mortality) are reported on the log scale; predictions and residuals are in log-units of mortality.

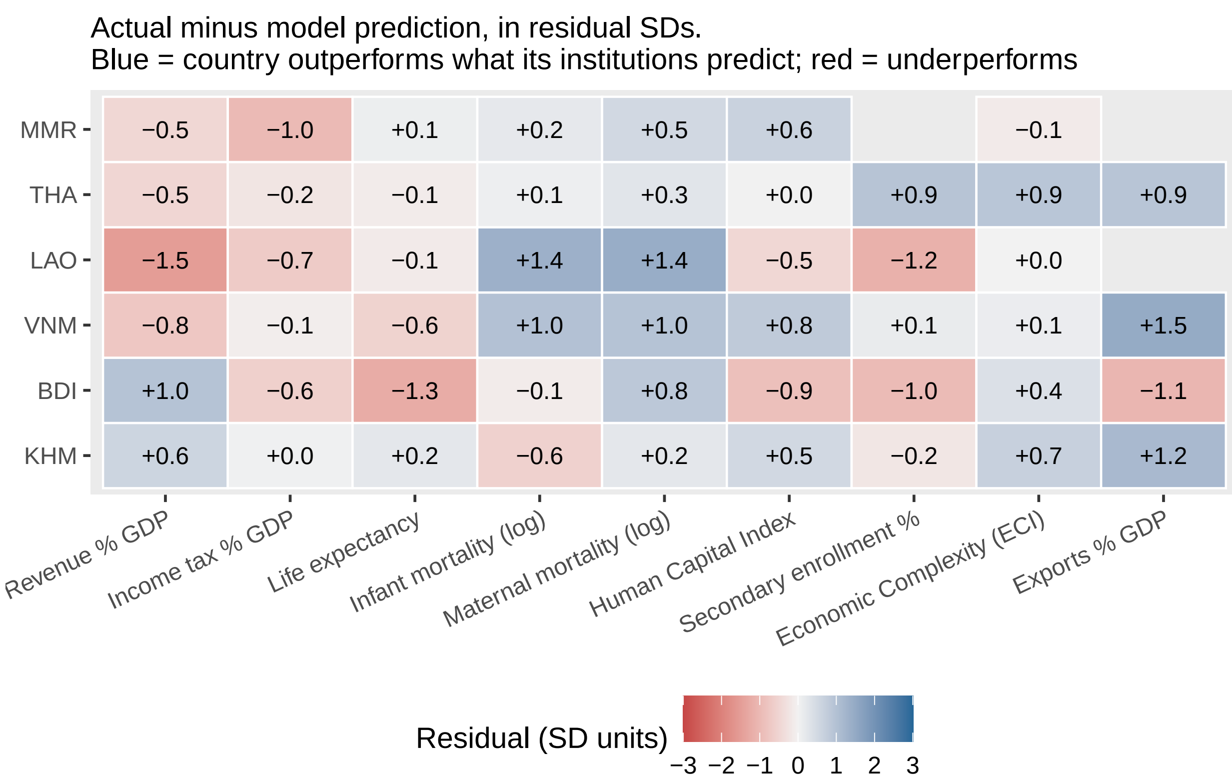


Table 9: Per-country actual vs predicted on each outcome. Residual SD > 0 means the country does better than its V-Dem fingerprint predicts; SD < 0 means worse.

Country	Bucket	Outcome	Actual	Predicted	Residual (raw)	Residual (SD)
KHM	Fiscal	Revenue % GDP	24.06	18.51	5.55	0.59
KHM	Fiscal	Income tax % GDP	4.14	4.01	0.13	0.04
KHM	Health	Life expectancy	70.67	69.67	1.00	0.23
KHM	Health	Infant mortality (log)	3.01	3.34	-0.33	-0.58
KHM	Health	Maternal mortality (log)	4.92	4.70	0.22	0.23
KHM	Education	Human Capital Index	0.49	0.45	0.04	0.53
KHM	Education	Secondary enrollment %	60.06	64.24	-4.18	-0.22
KHM	Complexity	Economic Complexity (ECI)	-0.33	-0.85	0.52	0.69
KHM	Trade	Exports % GDP	71.36	35.43	35.94	1.16
BDI	Fiscal	Revenue % GDP	29.53	20.48	9.06	0.97
BDI	Fiscal	Income tax % GDP	3.18	5.05	-1.87	-0.61
BDI	Health	Life expectancy	63.65	69.20	-5.55	-1.25
BDI	Health	Infant mortality (log)	3.45	3.52	-0.07	-0.13
BDI	Health	Maternal mortality (log)	5.97	5.17	0.81	0.85
BDI	Education	Human Capital Index	0.39	0.45	-0.06	-0.89
BDI	Education	Secondary enrollment %	46.39	65.02	-18.62	-0.98
BDI	Complexity	Economic Complexity (ECI)	-0.68	-0.95	0.28	0.37
BDI	Trade	Exports % GDP	5.29	38.20	-32.91	-1.07
VNM	Fiscal	Revenue % GDP	18.78	25.89	-7.11	-0.76
VNM	Fiscal	Income tax % GDP	5.38	5.68	-0.30	-0.10
VNM	Health	Life expectancy	74.59	77.07	-2.48	-0.56
VNM	Health	Infant mortality (log)	2.64	2.06	0.58	1.00
VNM	Health	Maternal mortality (log)	3.87	2.96	0.91	0.96
VNM	Education	Human Capital Index	0.69	0.63	0.06	0.82
VNM	Education	Secondary enrollment %	97.25	94.41	2.84	0.15
VNM	Complexity	Economic Complexity (ECI)	0.35	0.27	0.08	0.11
VNM	Trade	Exports % GDP	86.47	41.15	45.32	1.47
LAO	Fiscal	Revenue % GDP	15.12	29.35	-14.23	-1.52
LAO	Fiscal	Income tax % GDP	3.35	5.50	-2.15	-0.70
LAO	Health	Life expectancy	68.96	69.59	-0.63	-0.14
LAO	Health	Infant mortality (log)	3.56	2.78	0.78	1.35
LAO	Health	Maternal mortality (log)	4.72	3.37	1.35	1.43
LAO	Education	Human Capital Index	0.46	0.49	-0.03	-0.49
LAO	Education	Secondary enrollment %	54.50	76.75	-22.25	-1.16
LAO	Complexity	Economic Complexity (ECI)	-0.58	-0.58	0.00	0.01
THA	Fiscal	Revenue % GDP	19.97	24.67	-4.71	-0.50
THA	Fiscal	Income tax % GDP	6.08	6.81	-0.73	-0.24
THA	Health	Life expectancy	76.41	76.98	-0.57	-0.13
THA	Health	Infant mortality (log)	2.08	2.04	0.04	0.07
THA	Health	Maternal mortality (log)	3.53	3.27	0.25	0.27
THA	Education	Human Capital Index	0.61	0.61	0.00	0.02
THA	Education	Secondary enrollment %	109.85	91.89	17.96	0.94
THA	Complexity	Economic Complexity (ECI)	0.89	0.21	0.68	0.90

Country	Bucket	Outcome	Actual	Predicted	Residual (raw)	Residual (SD)
THA	Trade	Exports % GDP	70.06	41.50	28.57	0.93
MMR	Fiscal	Revenue % GDP	14.48	18.98	-4.51	-0.48
MMR	Fiscal	Income tax % GDP	0.96	4.05	-3.08	-1.00
MMR	Health	Life expectancy	66.89	66.49	0.39	0.09
MMR	Health	Infant mortality (log)	3.53	3.42	0.11	0.19
MMR	Health	Maternal mortality (log)	5.22	4.72	0.50	0.52
MMR	Education	Human Capital Index	0.48	0.43	0.04	0.65
MMR	Complexity	Economic Complexity (ECI)	-0.98	-0.87	-0.11	-0.15

Do two PCs explain the outcomes?

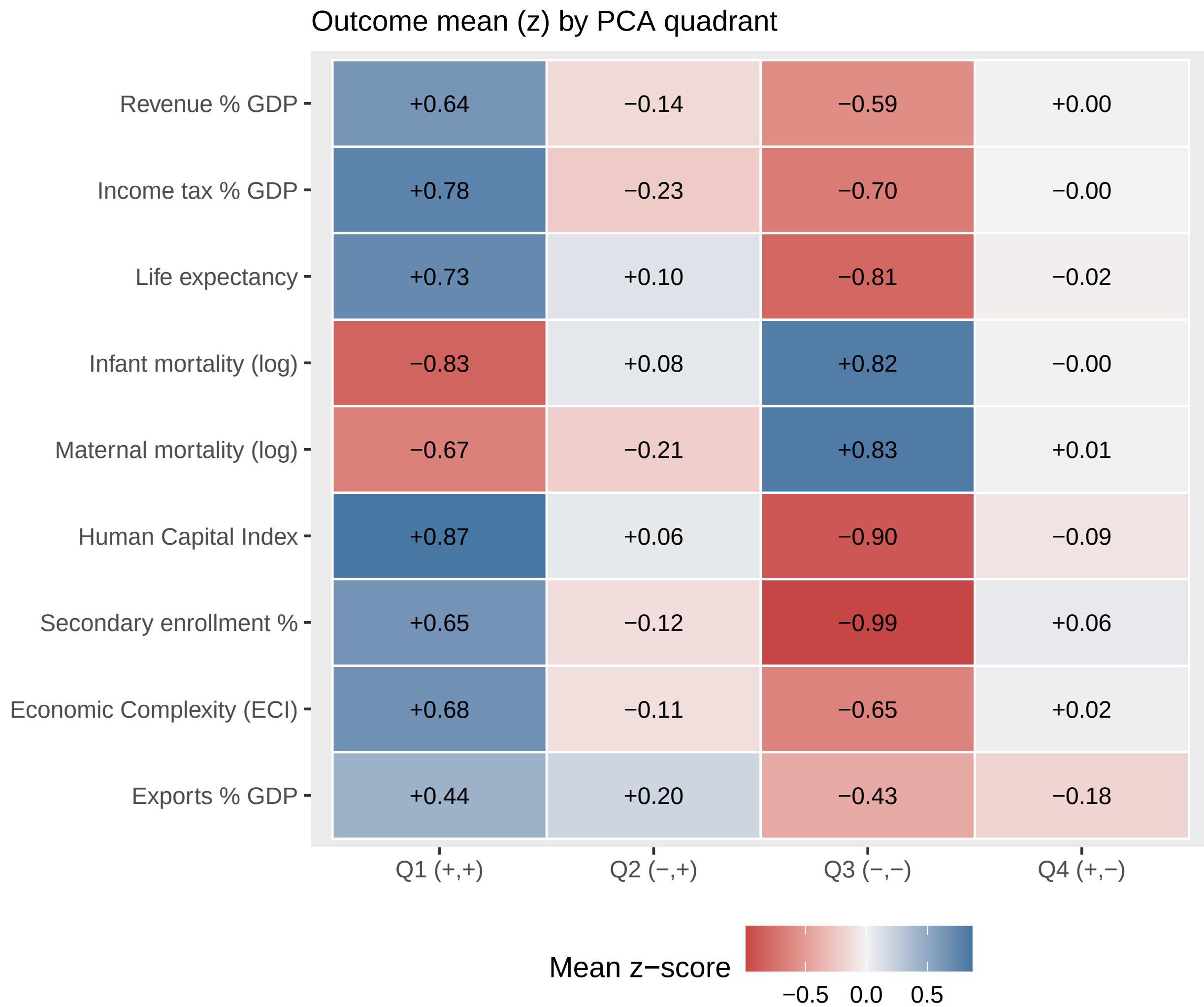
For each outcome we fit a simple OLS regression `y ~ PC1 + PC2 + PC1:PC2`. The interaction term is what makes this a quadrant story — a pure additive fit can't tell Q1 apart from "a bit of Q2 plus a bit of Q4." Outcomes are z-scored so coefficients are directly comparable across rows.

Table 10: OLS on z-scored outcomes. `PC1`, `PC2`, and `PC1:PC2` are standardized coefficients: how a one-standard-deviation move along each axis (or their product) shifts the outcome, in outcome-SD units. Compare to the best-depth XGBoost R^2 above to see how much signal lives outside the first two PCs.

bucket	label	n	r2	PC1	PC2	PC1:PC2
Complexity	Economic Complexity (ECI)	172	0.362	0.059	0.079	-0.001
Education	Human Capital Index	156	0.536	0.067	0.118	-0.002
Education	Secondary enrollment %	144	0.483	0.077	0.089	-0.006
Fiscal	Income tax % GDP	166	0.455	0.059	0.099	0.005
Fiscal	Revenue % GDP	170	0.297	0.054	0.067	0.000
Health	Infant mortality (log)	170	0.497	-0.066	-0.102	0.002
Health	Life expectancy	171	0.454	0.061	0.103	-0.003
Health	Maternal mortality (log)	170	0.436	-0.058	-0.103	0.004
Trade	Exports % GDP	155	0.186	0.020	0.114	0.004

Quadrant × outcome heatmap

Each cell is the mean z-score of an outcome among countries in that quadrant. Blue = outcome runs high in this quadrant, red = runs low, grey = near the sample average. Reading one row across the four quadrants is the quadrant-labeling exercise: it says which combinations of institutional arrangements are associated with better fiscal / health / education / complexity outcomes.



Quadrant count

Table 11: Countries per quadrant. Small-N quadrants make the row means above noisier.

quadrant	n
Q1 (+,+)	47
Q2 (-,+)	32
Q3 (-,-)	45
Q4 (+,-)	48

Top bundle per outcome

For each outcome, the highest-weight bundle from its best-depth model — the *set* of V-Dem features that the most-frequented tree paths repeatedly split on. Weight is the total leaf cover routed through paths that use exactly this set, so the top bundle is the “decision rule” the model uses most often, not necessarily the bundle with the strongest direct correlation to the outcome. Each feature is listed with its V-Dem codebook description for interpretability.

Revenue % GDP (Fiscal)

- **Best depth:** 5
- **OOS R^2:** 0.371
- **Bundle size:** 5
- **Bundle weight:** 106 (top 1 of 536 distinct bundles)

Features in this bundle:

- `v2dlconslt` — Range of consultation
- `v2elirreg` — Election other voting irregularities
- `v2elpaidig` — Election paid interest group media
- `v2smarrest` — Arrests for political content
- `v2strenarm` — Remuneration in the Armed Forces

Income tax % GDP (Fiscal)

- **Best depth:** 5
- **OOS R^2:** 0.581
- **Bundle size:** 4
- **Bundle weight:** 109 (top 1 of 917 distinct bundles)

Features in this bundle:

- `v2caviol` — Political violence
- `v2exdfcbhs` — HOS appoints cabinet in practice
- `v2jupoatck` — Government attacks on judiciary
- `v2smprivex` — Privacy protection by law exists

Life expectancy (Health)

- **Best depth:** 3
- **OOS R^2:** 0.629
- **Bundle size:** 3
- **Bundle weight:** 170 (top 1 of 726 distinct bundles)

Features in this bundle:

- `v2castate` — Engagement in state-administered mass organizations
- `v2excrptps` — Public sector corrupt exchanges
- `v2mefemjrn` — Female journalists

Infant mortality (log) (Health)

- **Best depth:** 5
- **OOS R^2:** 0.71
- **Bundle size:** 1
- **Bundle weight:** 782 (top 1 of 3381 distinct bundles)

Features in this bundle:

- `v2elsnlsff` — Subnational election unevenness

Maternal mortality (log) (Health)

- **Best depth:** 3
- **OOS R^2:** 0.684
- **Bundle size:** 3
- **Bundle weight:** 206 (top 1 of 992 distinct bundles)

Features in this bundle:

- `v2elvotbuy` — Election vote buying
- `v2pehealth` — Health equality
- `v2smfordom` — Foreign governments dissemination of false information

Human Capital Index (Education)

- **Best depth:** 3
- **OOS R^2:** 0.771
- **Bundle size:** 3
- **Bundle weight:** 220 (top 1 of 847 distinct bundles)

Features in this bundle:

- `v2cldmovem` — Freedom of domestic movement for men
- `v2lgsrvlo` — Lower chamber members serve in government
- `v2smpolcap` — Political parties cyber security capacity

Secondary enrollment % (Education)

- **Best depth:** 4
- **OOS R^2:** 0.517
- **Bundle size:** 4
- **Bundle weight:** 98 (top 1 of 811 distinct bundles)

Features in this bundle:

- `v2catrauni` — Engagement in independent trade unions
- `v2clrgunev` — Subnational civil liberties unevenness
- `v2pepwrort` — Power distributed by sexual orientation
- `v2pscnslnl` — Candidate selection-national/local

Economic Complexity (ECI) (Complexity)

- **Best depth:** 3
- **OOS R^2:** 0.45
- **Bundle size:** 3
- **Bundle weight:** 146 (top 1 of 483 distinct bundles)

Features in this bundle:

- `v2jucorrdc` — Judicial corruption decision
- `v2meslfcen` — Media self-censorship
- `v2smgovsmalt` — Government social media alternatives

Exports % GDP (Trade)

- **Best depth:** 6
- **OOS R^2:** 0.155
- **Bundle size:** 2
- **Bundle weight:** 75 (top 1 of 205 distinct bundles)

Features in this bundle:

- `v2peedueq` — Educational equality
- `v2smfordom` — Foreign governments dissemination of false information

Notes and caveats

- **Cross-sectional.** Everything here is a single-year snapshot. No causal claims.
- **Transforms.** Infant and maternal mortality are modeled in logs. SHAP units for those outcomes are log-units of mortality.
- **Regularisation.** `min_child_weight = 5` keeps leaves from specialising on one or two countries with N in the 130-190 range. CV-tuned `nrounds` plus the depth sweep are the only moving pieces.
- **Bundle extraction.** A bundle is the *set* of distinct V-Dem features used on any split along the path from root to a given leaf. Weight sums leaf `Cover` (approximately the number of training samples reaching that leaf), aggregated across all leaves in all trees whose feature-set matches. Bundles are direction-free — they say "these features co-occur in decisions" rather than "high A + low B co-occur."
- **Feature overlap.** V-Dem indicators are highly collinear. When two similar features (e.g. two measures of executive respect for law) show up together in a bundle, it may mean the model is using one as a noisy proxy for the other rather than that they genuinely interact. Treat top bundles as hypotheses.